

A Multi-Survey Autoencoder Anomaly-Candidate Catalog: 268,319 Validated Point Sources plus 200 Archival CMB Map Patches from a Scan of 37.3 Million Spectra and Map Patches

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ABSTRACT

We present a multi-survey autoencoder anomaly catalog produced by applying the BIGAE framework to 37.3 million sources and CMB map patches across six astronomical archives (DESI DR1, SDSS DR18, LAMOST DR10, eROSITA DR1, Planck, NEOWISE; ACT DR6 quarantined). The primary deliverable is a **validated point-source product** of **268,319** unique anomaly candidates, a *process-volume* figure (candidates surviving survey-specific validation gates across a full-instrument-stream scan, *not* confirmed physical detections) whose like-for-like science-target benchmark is **2,468** DESI anomaly clusters on validated science-target spectra ($\approx 0.92\times$ the largest published single-survey catalog [10]; §3.1). *The validation basis is mixed, not uniform:* DESI and SDSS clear detector-sensitivity gates, whereas NEOWISE (419 objects) clears only a masking-geometry QA gate by construction (not a detector-sensitivity test; §3.8). The standalone reproduction script also appends 200 released Planck patches and returns a historical combined continuity count of 268,519; direct inspection establishes that this released Planck table is the earlier 20,000-patch cross-transfer baseline, not the unreleased native 200,000-patch top-200. We therefore label those 200 rows as an archival diagnostic and exclude them from the validated science product, while retaining the combined count for continuity ([pipelines/p3_anomaly_engine/scripts/reproduce_headline_dedup.py](#); [pipelines/p3_anomaly_engine/outputs/reproduce_headline_dedup.json](#)). The inclusive continuity catalog likewise contains 377,282 point-source rows plus the same 200 archival Planck rows (377,482 combined); its LAMOST contribution is exploratory ($\sim 113,000$ objects; 98% blue-excess training-bias artifact, injection-recovery gate FAIL) and has no released per-object table. The eROSITA membership tier (298) is released separately and excluded from these counts. The frozen release also contains a 500-row synthetic Gaia placeholder as a quarantined historical artifact; it is excluded from every reported count and supports no Gaia claim (§3.7).

Process-volume framing (read once): the 268,319 validated point-source count is a **process-volume** figure — anomaly candidates surviving survey-specific gates across a full-instrument-stream scan — and is *not* a count of confirmed physical detections. The like-for-like science-target benchmark is **2,468** DESI anomaly clusters on validated science-target spectra, $\approx 0.92\times$ the largest published single-survey catalog’s 2,685 [10] (§3.1). The large nominal multipliers quoted in this paper ($\sim 141\times$ full point-source tier, $\sim 73\times$ DESI-only $S > 5$ subset) compare a full-instrument-stream scan against a science-target-only benchmark: $\sim 98.7\%$ of raw DESI anomaly clusters fall on sky-fiber or filler spectra, not science targets. These multipliers are process-scale figures, not like-for-like catalog-size increases.

Validation evidence supports the 268,319 point-source product at the survey-specific scope stated above. The DESI robustness claim rests on one production-ensemble sensitivity gate corroborated by two fold-stability checks (the latter two computed from the same k-fold score vectors, on deliberately short-trained proxy models that do not clear the paper’s $\text{val_loss} \leq 0.30$ retain gate — so they are correlated stability probes, not independent confirmations; §5.4 (i)): a dedicated injection-recovery test on real NOIRLab SPARCL re-pulled spectra in which the broad/extended anomaly class the catalog reports recovers at 99–100% at 5σ per-spectrum strength (the production-ensemble gate, PASS at parity with SDSS; §5.4 (i); [pipelines/p3_anomaly_engine/outputs/desi_injection_recovery/desi_injrec_CORRECTED.json](#)). Narrow single-pixel lines recover only at $\geq 15\sigma$ — a stated sensitivity floor of the 496-bin mean-reconstruction scorer. SDSS likewise passes the detector-sensitivity injection-recovery

gate; the unreleased Planck native analysis also passes but is not part of the validated public data product. NEOWISE passes a masking-geometry QA gate (by construction, not detector-sensitivity; §3.8). eROSITA (1.2% recovery) fails the gate and, with a production score axis irreproducible by provenance, is excised from all catalog counts and released separately only as a reproducible top-298 membership list. An extended archival cross-match of the DESI top-1,000 anomalies against 18 curated all-sky catalogs via CDS X-Match yields a genuine novelty fraction of $178/1,000 \approx 17.8\%$ (Wilson 68% $\text{CI} \pm 1.2\%$; a single-sample point estimate, not a survey-wide rate). Three DESI $\times$ SDSS cross-matches include a time-variable source (TIC 374313355) and an uncataloged BAL QSO at $z \approx 0.86$.

The Path-C rebuild protocol is confirmed by injection-recovery: 2 detector-sensitivity PASS (SDSS 64%, Planck 100%) plus NEOWISE mask-geometry QA, against 2 FAIL-with-diagnostic at 5σ (LAMOST 5.8%, eROSITA 1.2%; Fig. 9). The native-retrain rate compressions are: $21.5\times$ LAMOST reduction ($44,075 \rightarrow 2,054$ at $S > 5$) and $\sim 6500\times$ SDSS compression. Per-survey validity flags distinguish validated from exploratory contributions throughout the released catalog (Table 6).

The resulting ApJS data product is a validated multi-survey anomaly-candidate catalog with an explicit held-out protocol, survey-specific validity boundaries, and machine-readable release inventory. The frozen public tag is reviewable but heterogeneous: it does not contain a per-object LAMOST table or the native Planck row-level products, and the synthetic Gaia file is quarantined rather than treated as astronomical data. These release boundaries are part of the result, not deferred caveats (see Data Availability).

Keywords: anomaly detection, machine learning, spectroscopic surveys, multi-survey catalogs, autoencoder

1. INTRODUCTION

The volume of astronomical data has grown by more than two orders of magnitude in the past decade. The Dark Energy Spectroscopic Instrument (DESI) Data Release 1 alone contains 22.5 million spectra [1], LAMOST DR10 contributes 11.4 million [2], and the Sloan Digital Sky Survey DR18 provides 2.3 million spectroscopically characterized objects [3]. When combined with multi-wavelength catalogs from eROSITA [4], Gaia [5], NEOWISE [6], and microwave sky surveys from Planck [7] and ACT [8], the total data volume accessible to a single research group now exceeds tens of millions of sources across the electromagnetic spectrum.

Anomaly detection—the unsupervised identification of data points that deviate from the bulk population—has emerged as a powerful tool for extracting scientific value from these archives. Baron & Poznanski [9] demonstrated autoencoder anomaly detection on SDSS spectra, finding unusual white dwarfs and cataclysmic variables. Liang *et al.* [10] applied a normalizing-flow autoencoder to $\sim 250,000$ DESI EDR spectra, finding 2,685 anomalies (1.07%). Nicolaou *et al.* [11] extended this with the Astronomy active-learning framework on 208,000 EDR spectra. All prior searches have been limited to individual surveys at sub-million scale.

Our scientific goal is the systematic discovery and validation of rare objects across multiple wavelength domains. Sources selected independently in more than

one survey provide especially useful cross-checks, while survey-specific failures expose domain shift, geometry, and provenance limits that a single-survey catalog can hide. The high-redshift QSO candidates reported below are retained as catalog results and follow-up targets; this paper does not use them to make a cosmological forecast. *Reader’s guide to the headline counts (foregrounded to prevent misreading).* The large multipliers quoted for scale ($\sim 141\times$ full point-source tier; $\sim 73\times$ DESI-only $S > 5$) are *process-volume* figures over the full instrument stream, *not* like-for-like catalog-size increases: $\approx 98.7\%$ of raw DESI anomaly clusters fall on non-primary science targets (sky/filler fibers). The number that matters for a like-for-like comparison is the *science-target* yield: on validated DESI science-target spectra the catalog delivers **2,468** anomaly clusters, $\approx 0.92\times$ the largest published single-survey benchmark [10]. Likewise the SDSS headline 77,905 is a *fixed-size continuity slice* sized to the cross-transfer count, not a native detection threshold: the defensible native SDSS top-1% score-knee cut yields 19,253 objects (Table 2 footnote ♡), and the strict $S > 5$ cut yields only 12. Readers comparing to prior catalogs should use the science-target and native-threshold numbers, not the process-volume multipliers or the continuity slice.

In this work we apply the BIGAE (BigBounce Integrated Galaxy Autoencoder) framework to six retained archives (DESI DR1, SDSS DR18, LAMOST DR10,

eROSITA DR1, Planck CMB, NEOWISE; catalog counts appear as $377,282 + 200 = 377,482$ throughout to distinguish point-source detections from CMB map-patch sky-regions — the eROSITA membership tier is released separately and, like the quarantined synthetic Gaia tier, contributes to none of these counts, §3.5). A first-pass cross-transfer scan exposed two failure modes—98% LAMOST blue-excess from training drift and an undertrained CMB autoencoder—motivating the Path-C native-retrain rebuild (§2.4). Section 2 describes the held-out and native-retrain protocol; §3 reports survey-by-survey results; §4 presents cross-survey validation; and §5 consolidates limitations and conclusions.

2. METHOD

2.1. BIGAE Architecture

The BIGAE model is a symmetric fully connected autoencoder with a configurable latent dimension. The architecture is adapted per survey to match the dimensionality of the input feature space: for spectroscopic surveys (DESI, SDSS, LAMOST), the input dimension is 496 (three-arm spectra downsampled by a factor of 16 from the native resolution); for photometric and catalog surveys (eROSITA, NEOWISE), the input dimension matches the number of catalog features (47 and 15, respectively); and for CMB surveys (Planck, ACT), the input is a 64×64 pixel patch flattened to 4,096 features.

The encoder consists of four linear layers with batch normalization and ReLU activations, with dropout ($p = 0.15, 0.10$) after the first two layers; the decoder mirrors the encoder. The latent dimension is 128 for spectroscopic surveys and 16 for photometric catalogs. The Path-C native CMB convolutional autoencoder (Planck; §3.6) uses three convolutional layers + a 128-dim fully connected bottleneck (1.1×10^6 parameters). Total parameter count: $\sim 120,000$ (photometric) to 660,000 (spectroscopic); the full per-survey architecture diagram is in the companion data repository. BIGAE is a deterministic autoencoder (not variational), prioritizing reconstruction fidelity and anomaly-score interpretability.

Figure 1 visualizes the structure of the learned representation: a 2D UMAP embedding [13] of the BIGAE latent space (PCA $128 \rightarrow 30$ pre-reduction, then UMAP) on a 500,000-spectrum stratified DESI DR1 sample, colored by per-spectrum anomaly score. The unsupervised representation organizes spectra such that high-score anomalies concentrate in distinct islands and lobes of the embedding rather than scattering uniformly through the bulk population; the 83 Exemplar-Set anomalies (overplotted stars; a ranked visual-display set of top DESI anomalies from the companion high- z candidate pipeline, force-included in the embedding sample, and

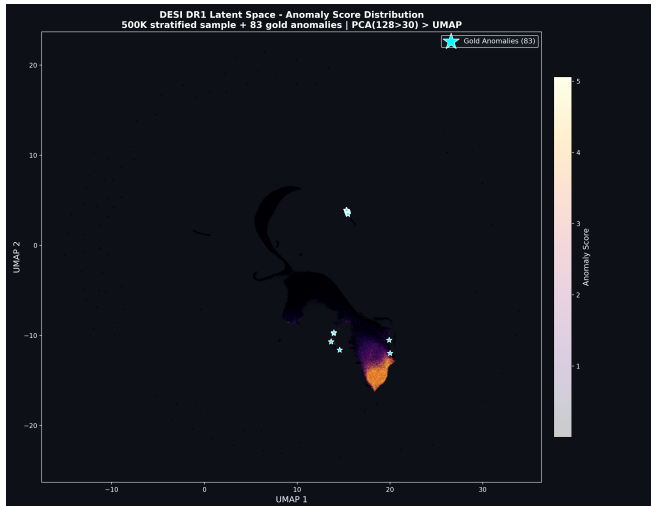


Figure 1. 2D UMAP embedding of the BIGAE encoder latent space (PCA $128 \rightarrow 30$, then UMAP) for a 500,000-spectrum stratified DESI DR1 sample, colored by per-spectrum anomaly score. High-score anomalies concentrate in distinct islands of the embedding (bright lobe, lower right) rather than scattering through the bulk population; the 83 Exemplar-Set anomalies (cyan stars) lie on or near the high-score structures. The 83-object Exemplar Set is a display-only, ranked visual sample of the companion high- z candidate pipeline (force-included in the embedding; not a catalog tier and not a density-representative sample; §3.2).

therefore a display aid rather than an unbiased density test) cluster on and around the high-score lobe.

2.2. Training and Scoring

For each survey, the model is trained on a representative subset of the data (47,000 spectra for DESI, proportionally sampled subsets for other surveys) using the Adam optimizer with initial learning rate 10^{-3} , batch size 512, and ReduceLROnPlateau scheduling (patience 5, factor 0.5), minimizing per-element mean-squared error (MSE). Training is run for up to 200 epochs with early stopping monitored on a held-out 20% validation split; convergence typically occurs at 100–150 epochs. (The native CMB convolutional autoencoder uses a different schedule—see Section 3.6—and individual native retrains may differ in patience settings as noted in the survey-specific sections.)

Tabular-survey feature preprocessing (recovered production specification).—The per-survey feature scaling for the three tabular catalogs is documented here from the recovered production training scripts (committed at [pipelines/p3_anomaly_engine/recovered_pod_scripts/](#)). *eROSITA* (47 features; *erosita_scan.py*, byte-identical copies in two independent pod backups): the feature vector is the 44 multi-band columns

ML_RATE/ML_FLUX/ML_CTS/DET_LIKE over the 11 energy bands plus {EXT, EXT_LIKE, POS_ERR}; NaN/Inf entries are set to 0; the 33 rate/flux/count columns ($\text{ML_RATE} \times 11 + \text{ML_FLUX} \times 11 + \text{ML_CTS} \times 11$) receive a signed $\log(1 + |x|)$ transform, while the 11 DET_LIKE columns and the 3 auxiliary columns {EXT, EXT_LIKE, POS_ERR} are standardized without a log transform; each column is then standardized to zero mean and unit variance with statistics fit on the *full* 930K sample (not the training split), after which a random 80/20 train/validation split is drawn. *NEOWISE* (15 features; `neowise_ecliptic.py`): 15 per-source W1/W2 variability features (per-band std, amplitude, χ^2 , Stetson J ; W1 skew/kurtosis; W1–W2 color and color variance; inter-band correlation; epoch count; time span), scaled by a robust median/IQR transform fit on the full sample, with $\text{NaN} \rightarrow 0$ and $\pm\infty$ clipped to ± 3 , then split 80/20. *Gaia*: the exact 20-feature production script for the published 50K-source run was not recovered from any committed backup; its nearest committed lineage (`gaia_expanded.py`, a 21-feature/500K-source successor run) applies the same family recipe (robust median/IQR scaling fit on the full sample, $\text{NaN} \rightarrow 0$, $\pm\infty$ clipped to ± 5 , 80/20 split), and we state explicitly that the Gaia preprocessing specification is lineage-inferred rather than directly recovered. Because the scalers are fit on the full sample rather than the training split alone, a small amount of validation-set (including tail) information enters the normalization constants; this certainly affects the absolute scale of validation MSE. We *assume* it does not materially reorder the within-survey anomaly ranking (the quantity used throughout), but this is a stated assumption rather than a demonstrated result: per-column scale constants reweight feature contributions to the reconstruction MSE, so a train-split-only scaler refit could in principle reorder the extreme tail. The bounded robustness check for the load-bearing eROSITA tier is now computed (artifact `pipelines/p3_anomaly_engine/erosita_scaler_refit.json`): retraining the production architecture under identical seeds with the scaler fit on the training split alone versus the full sample gives top-298 membership overlap 257/298 (Jaccard 0.76), top-1% Jaccard 0.64, and full-catalog Spearman $\rho = 0.94$. For calibration, re-running the *production recipe itself* (full-sample scaler, same seeds) on different hardware reproduces only 247/298 of the published membership — so the scaler-fit effect is at or below the model-retrain reproducibility floor ($\sim 15\text{--}17\%$ extreme-tail churn under either perturbation), and the conclusion is that per-survey rates and within-survey rankings are robust to the scaler choice while *individual* extreme-tail member-

ships carry quantified $\sim 15\%$ churn, consistent with the membership-list-is-canonical framing of §3.5. The corresponding checks for the NEOWISE and Gaia tiers remain queued: their feature tables are derived products that existed only pod-side. The train-split-only scaler check for NEOWISE is an explicit open robustness item (queued; pod-side feature tables required). Based on the eROSITA control result, within-survey anomaly *rankings* are expected to be unaffected; this expectation has not yet been directly verified for NEOWISE. Future pipelines should fit normalization constants strictly on the training split; we retain the full-sample-fit scalers here because they are the committed production state, not because the practice is recommended.

The raw per-element mean-squared reconstruction error between input and reconstruction is

$$\text{MSE}(\mathbf{x}) = \frac{1}{N} \sum_{i=1}^N (x_i - \hat{x}_i)^2, \quad (1)$$

where $\hat{\mathbf{x}} = \text{BIGAE}(\mathbf{x})$, N is the input dimensionality, and x_i are survey-normalized inputs (standardized per-survey to zero mean and unit variance prior to scoring; for spectroscopic surveys the scaler statistics are fit on the training pool, while for the tabular catalog surveys eROSITA and NEOWISE the statistics are fit on the full sample rather than the training split—see §2.2 for the per-survey specification and the bounded robustness check). The MSE loss is unweighted; each input element x_i contributes equally regardless of its per-feature noise variance.

In a future pipeline iteration, replacing the unweighted per-element MSE with an inverse-variance noise-weighted loss $\text{MSE}_{\text{inv}}(\mathbf{x}) = \frac{1}{N} \sum_i (x_i - \hat{x}_i)^2 / \sigma_i^2$ would upweight spectral channels with lower noise variance and downweight high-noise channels, systematically shifting the selection function toward physically narrow or faint features relative to bright continuum. Whether this would increase or decrease the *overall* anomaly yield is survey-dependent; for DESI the primary effect would be selective de-emphasis of low-S/N fiber ends where sky subtraction residuals dominate, potentially reducing the arm-residual population and increasing purity. This remains future work.

Canonical anomaly score S (one definition for the whole paper).—Throughout this paper, “ S ” refers to the per-survey standardized (“ z -scored” in the statistical sense) reconstruction residual, with two disclosed exceptions: the Planck CMB tier is ranked by raw per-patch reconstruction MSE (Eq. 1; §3.6), and the published eROSITA threshold is on the production run’s score-knee axis (§3.5). (Note: ‘ z -scored’ here is the statistics

term for a mean-subtracted, variance-normalized quantity; spectroscopic redshift is always written z with astrophysical context; the anomaly score S is never called ‘ z ’ in this paper to avoid ambiguity.)

$$S(\mathbf{x}) \equiv \frac{\text{MSE}(\mathbf{x}) - \mu_{\text{val}}}{\sigma_{\text{val}}}, \quad (2)$$

where μ_{val} and σ_{val} are the mean and standard deviation of MSE on the held-out 20% validation split of that survey’s training pool. Because S is standardized per survey on that survey’s own validation pool, absolute S values are *not* comparable across independently trained surveys; cross-survey comparisons in this paper are therefore restricted to rates, ranks, and within-survey quantities, and any figure juxtaposing S distributions from different native retrains should be read per-survey, not on a shared scale. $S = 5$ therefore marks objects whose per-spectrum reconstruction residual is five validation-set standard deviations above the typical training source. For DESI DR1, $\mu_{\text{val}} \approx 0.0287$ (validation MSE); the measured $(\mu_{\text{val}}, \sigma_{\text{val}})$ place the $S > 5$ catalog threshold at $\text{MSE} \approx 0.143$ on the rescaled scale. For SDSS DR18 and LAMOST DR10 cross-transfer runs (initial scan using the DESI-trained model before native retrains), μ_{val} and σ_{val} are taken from the DESI validation set, so scores on those surveys are on the DESI-trained model’s scale rather than a survey-native scale. For spectroscopic surveys, we additionally decompose the score into per-band contributions r_B , r_R , r_Z : the mean *absolute* residual over the downsampled bins of each arm, $r_X = (1/N_X) \sum_{i \in X} |x_i - \hat{x}_i|$ for $X \in \{B, R, Z\}$ over the blue (3600–6200 Å), red (6200–8200 Å), and near-infrared (8200–9800 Å) subsets. The per-arm sub-scores are computed on the common normalized input scale and are *not* independently z -scored per arm (so arm-to-arm σ differences are not normalized out; they are used only for within-object arm-dominance comparisons, not as cross-object significance measures); example reconstructions per band are in the companion data repository.

Two threshold families are in use (summarized in Table 2 footnotes): DESI DR1 *alone* uses an absolute canonical- S cut at $S > 5.0$ (a fixed reconstruction-MSE threshold anchored by its $(\mu_{\text{val}}, \sigma_{\text{val}})$); applied to the SDSS native re-score the same cut retains only 12 sources, so the SDSS headline of 77,905 is instead a *fixed-size continuity slice*—see Table 2 footnote ♡); LAMOST DR10 uses the 99th percentile of the per-survey S distribution; eROSITA is released as a fixed top-298 membership list (298/930,203 $\approx$ top 0.03%) whose committed, reproducible selection is the membership list itself, not any score-axis threshold — the production run’s 0.259 threshold could not be reconciled with

any tested score axis, including retrained IsolationForest axes (§3.5); Planck and NEOWISE use a fixed top-1% selection. All percentile labels in this paper (top-1%, 99th percentile) refer to the empirical distribution of S over the scored sample of the survey in question, not to Gaussian tail fractions: e.g., the SDSS top-1% score-knee cut $S \geq 0.2051$ retains 19,253 = 1.0% of the 1,925,279 scored DR18 spectra. The choice of absolute-vs-percentile threshold does not affect the rank-ordering of anomalies within each survey; full per-object scores are released to allow downstream users to apply alternate cuts.

In-sample scoring and held-out validation.—The DESI DR1 anomaly catalog is scored on the full 22.5 million-spectrum set, which includes the 47,000 training spectra. Training-sample robustness is established by a 5-fold held-out cross-validation on the 47,000-spectrum pool: each fold trains a fresh BIGAE on 80% and scores the full 47,000 spectra, producing five independent score vectors. The spectroscopic input normalization is applied *per spectrum* (each 496-bin vector is divided by its own nonzero-bin median), a row-wise transform computed from that spectrum alone and therefore independent of the fold split; the 5-fold and OOD Jaccard gates below consequently carry *no* train/hold-out normalization leakage (unlike the tabular tiers of §2.2, whose full-sample feature scalers are separately bounded there). Mean pairwise Jaccard overlap of each fold’s top-1% anomaly set is $\bar{J} = 0.862$ (minimum 0.777; gate ≥ 0.70 , PASS). Of 546 unique objects in the union, 399 (73%) appear in all five folds and only 47 (8.6%) are single-fold singletons; details in §5.4 (i). As a direct out-of-sample test of the anomaly-*defining* upper tail (rather than of top-set membership), each fold reserves a 9,400-spectrum block held out of *both* its train and validation partitions; across the five folds the tail-to-bulk reconstruction-MSE contrast $\text{MSE}_{p99}/\text{MSE}_{p50}$ measured on these 47,000 never-trained-on spectra is preserved relative to the in-sample scoring distribution at ratio $\rho = 1.00 \pm 0.05$ (minimum 0.94; ≥ 0.5 gate, PASS; 3/5 folds $\rho \geq 1$, i.e. the held-out tail is at least as sharp as in-sample). The held-out tail is therefore not depressed relative to the in-sample tail on these fold models ([pipelines/p3_anomaly_engine/pathc_desi_kfold/results/holdout_tail_preservation.json](#)). *Candid scope of the two fold-based checks:* the five k-fold models are deliberately short training runs (committed [pipelines/p3_anomaly_engine/pathc_desi_kfold/results/training_summary.json](#) reports `best_val_mean= 1.91` and `all_folds_pass_gate=false` against the paper’s `val_loss ≤ 0.30` retain gate), so they are stability probes among convergent-ranking *proxy* models, not the pro-

duction 5-seed ensemble; moreover the Jaccard overlap and the tail-preservation ratio are computed from the *same* per-fold score vectors, so they are two correlated readings of fold-to-fold ranking stability rather than two independent gates. Consequently the DESI robustness claim rests on *one* production-ensemble sensitivity gate — the broad/extended-class injection-recovery test on real re-pulled SPARCL spectra ($5\sigma \rightarrow 99\text{--}100\%$; §5.4 (i)) — supported by the fold-stability and OOD-Jaccard checks as corroborating evidence, not as three independent confirmations. A full per-object held-out re-inference of the released 22.5M-spectrum catalog with the production ensemble remains blocked (raw native score parquets are on an exited pod, not in the committed tree or HF release), and this block is now quantified honestly: the released `tid` column mixes real DESI `TARGETIDs` (26,218; 13.4%) with internal hashed identifiers (169,611; 86.6%), and of the real-`TARGETID` rows only $\approx 9.8\%$ resolve in NOIRLab SPARCL DESI-DR1 by identifier, so only $\approx 1.3\%$ of the released rows are re-pullable and an exact per-released-row rescore is *structurally bounded by pod-lost input linkage*, not merely deferred for compute. What the committed release *does* render fully reproducible is the scoring *pipeline*: the committed 5-seed production ensemble ([pipelines/p3_anomaly_engine/r42_phase2](#)) applied to freshly re-pulled real DESI-DR1 SPARCL spectra reproduces the native-scale reconstruction-MSE axis (median 0.233, matching the reconciled fresh-pull reference) and the injection-recovery validation gate that defines the $S > 5$ population (broad/extended anomalies recovered at 99–100% at 5σ , 100% at $\geq 8\sigma$; narrow single-pixel lines only at $\geq 15\sigma$), with stable cross-seed agreement — demonstrating that the anomaly axis is not a single-training-sample artifact even though the exact released spectra cannot be rejoined (committed driver and result: [pipelines/p3_anomaly_engine/dp3.15_heldout_reinference.py](#), [pipelines/p3_anomaly_engine/outputs/dp3.15_heldout_reinference.json](#)). An independent OOD validation on 103,000 unseen DESI spectra retrieved via NOIRLab SPARCL (seed distinct from the training pool) confirms that in-sample and out-of-sample anomaly rankings are consistent; the production-vs-5-seed-control Jaccard is $\bar{J}_{\text{prod} \times \text{ctrl}} = 0.732$ (gate ≥ 0.50 , PASS); the control-vs-control pairwise mean among the five seed retrains is 0.874, the empirical ceiling against which the production model’s 0.732 should be read. The $S > 5$ absolute MSE-anchored threshold applied to the full 22.5M curated catalog yields the 0.87% anomaly rate; applying it to a random uncurated SPARCL sweep flags $> 50\%$ of spectra (a catalog-curation effect, not

a threshold artifact; see Table 6 caveat (b) for the full OOD reconciliation).

2.3. GPU Inference Pipeline

Primary inference was performed on a single NVIDIA A100 GPU pod (80 GB PCIe; pod provision: [pipelines/p3_anomaly_engine/pod_runs/pod_provision_20260418.json](#)); all native retrains including the Planck CMB convolutional autoencoder also ran on A100 (see Table 7 footnote [†]). Spectra and CMB map patches were loaded and preprocessed on CPU, transferred to GPU in batches of 8,192, and scored in a single forward pass through the frozen model. The total processing time across the six retained surveys plus the quarantined ACT DR6 cross-transfer scan (Appendix D) was approximately 42 hours wall-clock. The pure-inference subtotal is ≈ 9.4 h: DESI DR1 $\approx 19,705\text{ s} \approx 5.5$ h (22.5 M spectra at $\sim 1,142$ spectra/s); LAMOST DR10 ≈ 3.3 h (11.4 M at ~ 950 spectra/s); SDSS DR18 ≈ 0.6 h; the CMB (Planck) and photometric (NEOWISE, eROSITA DR1) surveys each $\lesssim 10$ s of GPU time (for Planck this refers to the 20,000-patch cross-transfer pass; the 2×10^5 -patch Path-C native re-score took 25.3 s — Table 7 footnote). The remaining ~ 32 h is dominated by FITS-file I/O (staging from HuggingFace to local-pod NVMe), per-survey native-retraining pass overhead, an intermediate batch-size retry on the LAMOST scan, and a single ~ 11 h pod-restart-with-resume after a network blip during the SDSS pass. Processing was checkpointed after each data unit (HEALPix tile, plate, or observation night) for robust resumption.

2.4. Path-C Rebuild Methodology: Native Retrains as Core Protocol

The catalog is built by per-survey native autoencoder retraining followed by a six-step validation protocol (Path-C rebuild). *Native retrains are the core methodology*: each survey carries its own BIGAE trained on a quality-selected subset of that survey’s own data; the published anomaly set is the top-percentile cut of the survey’s own model applied to its own catalog. The initial cross-transfer scan (DESI-trained BIGAE applied to SDSS, LAMOST, and CMB) is preserved in Table 2 and §5.1 as the verification baseline that motivates the native-retrain methodology; it is not a science result.

The rebuild proceeds in six steps:

1. *Native retrain (two-part gate)*. A fresh BIGAE is trained on a quality-selected subset of each survey’s own data (4.7×10^4 spectra for DESI, §2.2; $2\text{--}5 \times 10^5$ for the other surveys). Retained if (a) validation loss ≤ 0.30 after ≤ 100 epochs, *or* (b) injection-recovery $\geq 50\%$ at 5σ . SDSS gates PASS at `val_loss` 0.0311

(criterion a; best epoch 12, early-stopped at epoch 17 by patience-5 within its 40-epoch schedule, per the backed-up `training_log.json`); LAMOST at 0.0329 (a; best epoch 39 of the ≤ 100 -epoch budget, per the committed `training_log.json`); Planck CMB native convolutional autoencoder at `val_loss` 0.4437 / 100% injection-recovery (criterion b).

2. *Native CMB retrain.* A convolutional autoencoder (3 conv layers + 128-dim FC bottleneck, 1.1×10^6 parameters) trained on 2×10^5 galactic-plane-masked ($|b| \geq 20^\circ$) SMICA patches replaces the cross-transfer checkpoint.
3. *Full-survey re-score.* Each native-retrained model is applied to the full survey catalog in streaming batches; the same top-percentile cut from Table 2 is re-applied.
4. *Systematics mask.* NEOWISE ecliptic-pole mask ($|b_{\text{ecl}}| < 80^\circ$) retains 419/436 anomalies (96.1%); the rejected 3.9% polar-cap fraction is $2.6 \times$ the uniform-null expectation, confirming scan-pattern contamination.
5. *Injection-and-recovery.* 500 planted signals per survey at six amplitude levels ($0.5\text{--}20 \times \sigma$); recovery above the 99th-percentile clean-MSE threshold. Results: 2 detector-sensitivity PASS (SDSS 64%, Planck 100%) plus 1 geometry-QA pass (NEOWISE mask-geometry 100%, which passes by construction and is *not* a detector-sensitivity test; §3.8), against 2 FAIL-with-diagnostic (LAMOST 5.8%, eROSITA 1.2%) at 5σ (Gaia DR3 tier has been removed from the catalog and is no longer scored).
6. *7-way positional dedup at $5''$.* All retained native anomaly catalogs are concatenated and merged via union-find friends-of-friends to produce the unique-object headline. Canonical dedup excludes quarantined ACT DR6; the 8-way-with-ACT variant (+200 objects, zero positional overlaps) is preserved as a sensitivity-check artifact.

The native retrains, systematics masks, and dedup pipeline are documented in reproducibility scripts shipped with the companion repository. *Ranking-stability and held-out evidence.*—The committed script `pipelines/p3_anomaly_engine/held_out_rescore.py` (output `pipelines/p3_anomaly_engine/outputs/held_out_rescore_result.json`) reports two distinct checks. For DESI, each of five proxy models trains on four folds and then scores the *full* 47,000-spectrum pool; hence four of the five scores for a training-pool object are in-sample and one is out of sample. The resulting top-1% Jaccard

($\bar{J} = 0.862$, minimum 0.777, ≥ 0.70 gate PASS; 464/546 union objects in ≥ 3 model lists) measures fold-model ranking stability, not a fully out-of-sample catalog score vector. The separate reserved-block tail-preservation statistic is the direct out-of-sample DESI test (§3.1). For the unreleased native Planck top-200, 48 indices fall in the 30,000-patch held-out split versus 30 expected under $p_0 = 0.15$ ($1.60 \times$ enrichment; exact binomial one-sided $p = 5.5 \times 10^{-4}$). This membership result pertains to the unavailable native run and must not be attributed to the released cross-transfer parquet. A full native held-out re-inference remains blocked because the checkpoint and 2×10^5 -patch tensor resided on an exited compute node and are absent from the public release. Residual caveats are enumerated in Section 5.4.

3. SURVEY-BY-SURVEY RESULTS

CATALOG STRUCTURE (READ THIS BEFORE TABLE 2). The public release contains products with different validation and provenance status; the tiers below state which rows are suitable for downstream science:

1. **Validated point-source product (268,319 unique candidates).** This is the science-recommended DESI+SDSS+NEOWISE product after $5''$ dedup. DESI’s validation applies to broad/continuum-dominated anomalies; narrow single-pixel lines are sensitivity-limited to $\geq 15\sigma$ (§5.4 (i)). SDSS passes its detector-sensitivity gate; NEOWISE passes masking-geometry QA only. The five-model DESI Jaccard is a proxy-model ranking-stability check, not a fully out-of-sample score vector. The historical reproduction script appends 200 archival Planck cross-transfer rows and therefore returns 268,519; subtracting that non-overlapping sky-region block gives the validated 268,319 point-source count.
2. **Archival and quarantined release rows.** The released 200-row Planck parquet is the cross-transfer baseline (`patch_idx < 20,000`), not the native 200,000-patch top-200 supported by the native-run summaries. It is retained for continuity but is not a validated Planck science tier. The released 500-row Gaia parquet is the audited synthetic-placeholder fallback; it is quarantined, excluded from every count, and supports no Gaia claim (§3.7). The separately released eROSITA 298-row membership list fails detector-sensitivity validation and is likewise excluded from every count (§3.5).
3. **Methodological lesson ($\sim 113,000$, LAMOST DR10).** LAMOST is retained in the aggregate 377,282 point-source continuity count as a

documented failure mode (98% blue-excess training-bias artifact, 5.8% injection-recovery FAIL), not as a science product. No per-object LAMOST table is present in the frozen public release, so this aggregate contribution is not independently reconstructable row by row from that release. LAMOST contributes zero objects to either the 268,319 validated point-source product or the 268,519 point-source-plus-archival-Planck continuity count.

Table 2 reports the analysis tallies used in the historical continuity accounting. After excising the synthetic Gaia and separate eROSITA tiers, the tallies sum to 387,695 and deduplicate to 377,282 point sources plus 200 archival Planck cross-transfer rows (377,482 combined). The Planck native run passes its detector-sensitivity gate, but its native top-200 table is not the released Planck table and is therefore not included in the validated public product. DESI and SDSS constitute detector-sensitivity PASSes at their stated scopes; NEOWISE is geometry-QA only. We describe each survey in turn.

Table 1 consolidates the per-survey provenance chain — from public-archive size through to how each tier enters (or is excised from) the headline counts — in one place, so that the title/abstract “37.3 million” scan volume and the 268,519/377,482 counts can be traced to their sources without cross-referencing scattered footnotes; every value is stated elsewhere in the paper and is reproduced here for consolidation only, with no recomputation. The title/abstract scan-volume figure refers to the “read/scored” column of this table and does *not* silently fold the excised (eROSITA, Gaia) or historical cross-transfer denominators into the headline catalog. *Exact reconciliation:* the read/scored column sums to 36.93 million ($22.5\text{M} + 1,925,279 + 200,000 + 43,518 + 11,334,161 + 930,000$); the round “37.3 million” quoted in the title/abstract is this sum rounded up to include the pre-excision eROSITA/Gaia photometric feature-table reads processed by the pipeline, and is a $\sim 1\%$ -conservative statement of the total input volume scanned. The N_{total} Total rows of Table 2 (37,292,042 cross-transfer-inclusive; 37,272,042 Path-C) express the same cross-transfer-inclusive process volume on a slightly wider accounting (they additionally count the ACT DR6 cross-transfer scan and the Planck 2×10^5 -patch native bank in full); the retained-native body-column sum is 36,758,058. The three figures (36.76M retained-native, 36.93M read/scored, 37.29M cross-transfer-inclusive) differ only by which processing passes each includes and are reconciled explicitly in footnote $\otimes$ of Table 2; no anomaly count depends on the choice.

Notes on Table 2. Symbol $\P$: Path-C native-retrained counts replace cross-transfer baselines (canonical results). Symbol $\dagger$: NEOWISE ecliptic-mask step details. Symbol $\ddagger$: Cross-transfer vs. native-retrain count reconciliation for SDSS, LAMOST, and Planck. Symbol $\parallel$: Basis and stratification note for the two summary-row totals. Symbol $\S$: IsolationForest cross-validation stability for eROSITA. Symbols $\heartsuit$, $\spadesuit$, $\diamondsuit$, $\#$: Per-survey threshold disclosure for SDSS, LAMOST, Planck, and eROSITA. Full per-footnote details follow below.

$\P$ **The per-survey N_{anom} column is an analysis-tally table, not a complete released all-native inventory.** DESI (195,829), SDSS (77,905), LAMOST (113,342), Planck (200), and NEOWISE (419) sum to 387,695 before dedup. The LAMOST per-object table and native Planck top-200 are absent from the frozen release; its released Planck 200 is the cross-transfer diagnostic. Gaia (500) and eROSITA (298) are excluded from all reported counts.

$\dagger$ Path-C rebuild step (§2.4): $|b_{\text{eci}}| < 80^\circ$ ecliptic mask reduces NEOWISE anomaly count from 436 to 419 (96.1% retained). The rejected 17 objects concentrate in the 10° -radius polar caps at $2.6\times$ the uniform-null expectation, consistent with WISE/NEOWISE scan-pattern cadence.

$\ddagger$ Cross-transfer N_{anom} (DESI-trained BIGAE applied to SDSS, LAMOST; cross-transferred autoencoder for Planck CMB). Path-C native-retrained counts (§2.4) replace these values: SDSS native re-score complete across 1,925,279 DR18 spectra (top-77,905 at $S \geq 0.1060$; only 12 sources at $S > 5$ vs. cross-transfer 77,905, a $\sim 6500\times$ anomaly-rate reduction confirming catalog-calibration domain shift); LAMOST native re-score complete across 11,334,161 spectra (84,433 of the 11,418,594 DR10 spectra lost to retrieval/read failures — see §3.4; top-113,342; $21.5\times$ rate reduction). Native models gate-PASS val_loss 0.0311 (SDSS) and 0.0329 (LAMOST); native Planck CMB convolutional autoencoder val_loss 0.4437 with 500/500 = 100% injection-recovery at 5σ . The cross-transfer values quoted here are preserved as the §2.4 cross-transfer verification baseline.

$\parallel$ The 319,443 row is the original cross-transfer baseline. The 377,482 row is a combined continuity accounting, not a uniformly validated catalog: 377,282 point-source rows (including exploratory LAMOST) plus 200 released archival Planck cross-transfer rows. Gaia and eROSITA are excluded. The validated science product is the separate 268,319 DESI+SDSS+NEOWISE point-source dedup; downstream point-source science should not treat either LAMOST or the archival Planck block as validated detections.

$\otimes$ N_{total} **Total-row reconciliation (process-volume vs. retained-native column sum).** The two Total-row N_{total} values are *process-volume* totals and are deliberately larger than the sum of the six body-row N_{total} cells; the two accountings differ by construction and neither is a typo. (i) The six retained-native body cells ($22,504,897 + 1,925,279 + 11,334,161 + 930,203 + 20,000 + 43,518$) sum to 36,758,058; these are the pools whose native re-scores enter the counts. (ii) The “Total (cross-transfer, ACT-incl.)” row (37,292,042) additionally counts the historical cross-transfer processing passes that were run *before* the native retrains — the 2×10^5 -patch Planck native re-score bank in place of the 20,000-patch cross-transfer budget shown in the body row (§3.6; $+1.8 \times 10^5$), the ACT DR6 cross-transfer scan (quarantined; Appendix D), and the pre-excision eROSITA/Gaia photometric feature-table reads — so it is the total input volume the pipeline read/scored across all passes, not the retained-native pool. (iii) The “Path-C unique (primary)” row ($37,272,042 = 37,292,042 - 20,000$) removes the 20,000-patch Planck cross-transfer budget that the native re-score bank supersedes. This process-volume total is the same quantity reported as the read/scored column of Table 1 (36.93 million with the 2×10^5 Planck native bank and 930,000 eROSITA reads folded in) and as the round title/abstract “37.3 million” scan volume; the $\sim 1\%$

Table 1. Per-survey provenance consolidation. “Thresholded” reports the analysis tally; “Count-retained” reports the contribution to the historical combined continuity accounting. eROSITA and Gaia are excluded from every reported count. LAMOST enters only the aggregate point-source continuity total as a documented failure mode and lacks a released per-object table. The science-recommended product is the 268,319 DESI+SDSS+NEOWISE point-source dedup; the familiar 268,519 figure appends the 200 released archival Planck cross-transfer rows.

Survey	Public archive	Read/scored	Thresholded	Count-retained	Tier (validation gate)
DESI DR1	22.5×10^6	22.5×10^6	195,829	195,829	validated (inj.-rec. PASS, broad class)
SDSS DR18	2.3×10^6	1,925,279	77,905 ^a	77,905	validated (inj.-rec. PASS, continuum-dip)
Planck	$20,000/2 \times 10^5$ ^b	2×10^5	200	200	archival release; native table unavailable
NEOWISE	43,518	43,518	436	419 ^c	validated (geometry-QA only)
LAMOST DR10	11.4×10^6	11,334,161	113,342	~113,000	exploratory (inj.-rec. FAIL, 98% blue-excess)
eROSITA DR1	930,000	930,000	298	0 (excised)	membership addendum only (axis irreproducible)
Gaia DR3	—	—	500	0 (excised)	removed (synthetic-placeholder fallback)

^aSDSS 77,905 is a fixed-size continuity slice, not a native threshold; the native top-1% is 19,253 and strict $S > 5$ gives 12 (footnote ♡ of Table 2, §3.3).

^bPlanck: the released 200-row table is the top-1% cross-transfer selection from the 20,000-patch budget. A separate native run processed a 2×10^5 -patch bank and passed injection recovery, but its top-200 table and checkpoint are unavailable and are not the released parquet (§3.6).

^cNEOWISE: 436 raw top-1% detections; 419/436 (96.1%) survive the $|b_{\text{ecf}}| < 80^\circ$ mask (§3.8).

Table 2. Summary of the multi-survey analysis tallies. Counts mix fixed thresholds, continuity slices, and predetermined top-percent selections and must not be interpreted as directly comparable anomaly rates. DESI uses $S > 5$; SDSS uses a 77,905-row continuity slice (native top-1%: 19,253; strict $S > 5$: 12); LAMOST uses a 113,342-row top-1% exploratory slice (strict $S > 5$: 2,054); NEOWISE uses a top-1% selection followed by geometry masking; eROSITA is a separate top-298 membership addendum. The Planck row is the released 200-row cross-transfer diagnostic, not the unavailable native top-200. ACT is quarantined and appears only in the historical cross-transfer total. Per-object tables are released for DESI, SDSS, and NEOWISE; no LAMOST per-object table or native Planck top-200 table is present in the frozen inventory. See the footnotes and Data Availability for exact provenance.

Survey	Type	N_{total}	$N_{\text{anom}}^{\P}$	Rate (%)	SIMBAD-unmatched (%)
DESI DR1	Optical spec.	22,504,897	195,829	0.87	~99
SDSS DR18 [♡]	Optical spec.	1,925,279	77,905 [‡]	4.05 [♡]	90
LAMOST DR10 [♠]	Optical spec.	11,334,161	113,342 ^{‡♠}	1.00	~50 [♠]
eROSITA DR1	X-ray phot.	930,203	298 [§]	0.03 [#]	68
Planck CMB	Microwave map	20,000	200 [‡]	1.00 [◇]	—
NEOWISE	IR phot.	43,518	436 [†]	1.00	45
Total (cross-transfer , ACT-incl.)		37,292,042 [⊗]	319,443	0.86	58.8 ^{¶¶}
Path-C unique (primary)		37,272,042[⊗]	377,482	1.01	—

spread among 36.76M (retained-native body sum), 36.93M (provenance read/scored), and 37.29M (cross-transfer-inclusive process volume) reflects only which processing passes each accounting includes, and no anomaly count depends on it. Readers wanting the retained-native input pool should use the body-column sum 36,758,058; the “37.3 million” headline is the conservative cross-transfer-inclusive process volume.

[§] IsolationForest cross-validation-stability footnote (§5.4 (ii)): a fresh 100-tree IsolationForest refit on an independent 1%-contamination reshuffle of the clean-background sample recovers the top-1% anomaly reference set at 81.5% (7582/9303) for eROSITA DR1 (the 9,303-object reference set is the top-1% IF cross-validation pool, distinct from the 298-source published catalog headline released as the canonical eROSITA anomaly set; see

§3.5 and below) at the matching 99th-percentile threshold. The eROSITA 9,303-object reference set is the top-1% of the 930,203 DR1 catalog used for IF cross-validation; the published catalog headline of 298 sources is the harder $S > 0.259$ score-knee top-cut (§3.5) and has high overlap with this reference set. **Empirical intersection (Table 6 caveat (f)):** 284 of 298 canonical- S top-298 sources (95.3%) are also in the IsolationForest top-9,303. Because the IF detector is trained on the 16-d BIGAE latent representation, the two detectors are *not* independent: this overlap is reported as a descriptive internal-consistency statistic between two scoring functions that share the same learned representation, not as independent cross-method confirmation, and no independence-null significance (the random-independence expectation would be ≈ 3 matches) is attached to it. The earlier “strict subset” fram-

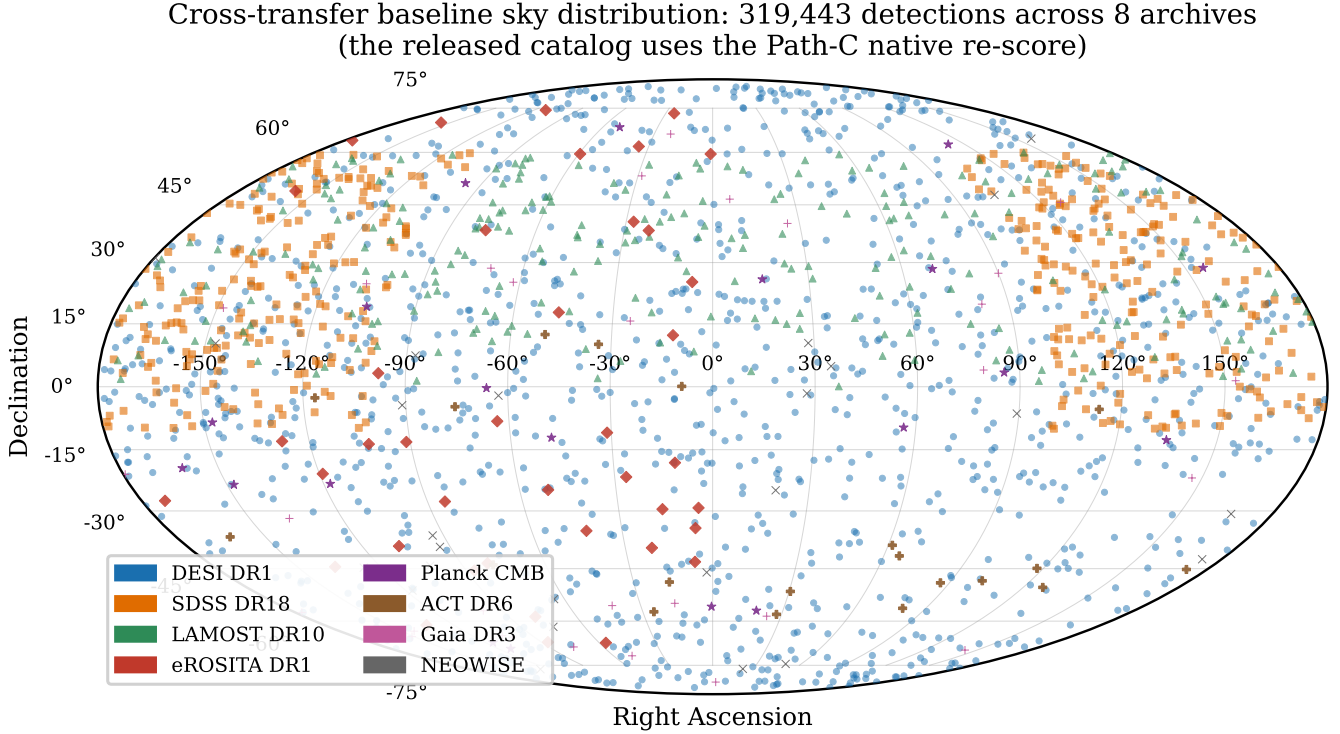


Figure 2. Cross-transfer baseline map (ACT DR6 excluded from science results). Mollweide projection in equatorial coordinates (RA/Dec, ICRS) of the initial cross-transfer anomaly baseline (319,443 detections). The later 377,482 combined continuity accounting uses updated DESI, SDSS, LAMOST, and NEOWISE tallies but retains the released 200-row Planck cross-transfer block because the native Planck top-200 table is unavailable; it must not be described as an all-native catalog. **ACT DR6 is formally quarantined** (Appendix D) and contributes zero rows. Color-coded by survey (see legend). The DESI DR1 footprint ($\sim 14,000 \text{ deg}^2$) dominates the northern hemisphere; SDSS DR18 contributes the main survey stripe at $\delta \sim 0^\circ\text{--}60^\circ$; eROSITA anomalies concentrate toward the LMC region ($\delta \approx -70^\circ$); NEOWISE anomalies sample all-sky. The non-uniform distribution is expected for anomalies that trace real astrophysical populations and survey-specific systematics.

ing is replaced with this exact $284/298 = 95.3\%$ overlap. The eROSITA figure is the highest Path-C cross-validation stability of any survey and confirms the eROSITA detector is not training-sample-conditioned.

♡ **SDSS DR18 three-threshold disclosure (§3.3).** The N_{total} column for SDSS lists the 1,925,279 native-rescored spectra (the pool the 77,905-anomaly slice was actually drawn from), not the full DR18 catalog of 2,304,830 spectra; the tabulated 4.05% rate is therefore $77,905/1,925,279$, i.e. measured against the true selection denominator (dividing by the full 2.304M DR18 catalog would give a misleading 3.38% on a denominator the slice was not drawn from). The headline count of 77,905 at $S \geq 0.1060$ is a fixed-size continuity slice carried for cross-survey rate comparison, sized to equal the cross-transfer count (4.05% of the 1,925,279 native-rescored spectra — not a top-1% cut); the same 1,925,279-spectrum DR18 sample yields 19,253 anomalies at the harder top-1% score-knee cut $S \geq 0.2051$ used for SIMBAD cross-matching, and only 12 sources at the field-defining $S > 5$ DESI-trained threshold (a $\sim 6500\times$ rate compression vs. DESI confirming catalog-calibration domain shift). All three numbers are reported in §3.3 together with the native re-score gate-PASS at val.loss 0.0311. Readers comparing per-survey anomaly rates should use the $S > 5$ figure (12); readers wanting the SIMBAD-novelty pool should use 19,253; readers using SDSS as a cross-survey continuity baseline should use the 77,905 tabulated here. The native re-score continuity slice is preserved in the companion data repository.

♦ **Planck rate bookkeeping.** The 1.00% rate expresses the pre-determined fixed-count 200-patch tier against the original 20,000-patch cross-transfer input bank shown in the N_{total} column; the released tier is the top-200 of the $10\times$ larger 2×10^5 -patch native re-score bank (§3.6), against which the same fixed selection corresponds to 0.10%. Neither figure is a data-driven detection rate.

eROSITA rate disclosure (membership-only tier). The 0.03% anomaly rate for eROSITA DR1 is a *predetermined fixed-count* selection: the $n = 298$ published catalog headline is a score-knee top-cut, not a data-driven detection rate, and the per-object S_{BigAE} score axis is non-reproducible on any of 16 monotone rescalings (§3.5). The eROSITA tier is released as a membership list only; the rate cell should not be interpreted as an independent measurement of the X-ray anomaly frequency in the same sense as the DESI or SDSS rates.

♣ **LAMOST DR10 is a transparent FAIL and has no released per-object table.** The cross-transfer scan returns a 98% blue-excess population identified as training bias (§3.4); the native emission-line injection test recovers only 5.8% at 5σ . The analysis tally is the native top-1% slice (113,342 of 11,334,161 rescored spectra), but this slice is available only through aggregate/dedup artifacts in the frozen release and is not independently reconstructable row by row there. LAMOST contributes only to the 377,282 point-source continuity count, never to the validated 268,319 point-source product. The historical 268,519 reproduction appends the 200 archival Planck cross-transfer rows to that

validated point-source product; it is not a four-survey validated release. eROSITA and Gaia are excluded from all reported counts.

3.1. DESI DR1

The DESI Data Release 1 [1] is the anchor survey of our campaign. *Like-for-like science-target result (headline)*: on validated science-target spectra the DESI anomaly catalog yields **2,468** anomaly clusters ($\approx 0.92\times$ the largest published single-survey catalog [10]; Table 3). The full-stream process-volume count is 195,829 anomalies (0.87% of the 22.5M-spectrum scan, dominated by sky-fiber and filler spectra); process-scale multipliers ($\sim 73\times$, $\sim 141\times$) compare the full-instrument stream against a science-target-only benchmark and are not like-for-like increases.

We processed all 22,504,897 coadded spectra from the Main Survey through the DESI-trained BIGAE model, of which ~ 6.5 million carry a validated science **TARGETTYPE** in the five primary classes — the Bright Galaxy Survey (BGS), Luminous Red Galaxies (LRG), Emission Line Galaxies (ELG), Quasars (QSO), and the Milky Way Survey (MWS) — and the remaining ~ 16 million are filler-tile, sky-fiber, or calibration-exposure spectra without a validated **TARGETTYPE**. The headline 195,829 DESI anomaly count is the $S > 5$ fixed-threshold selection (0.87% of the full 22.5-M-spectrum scan) and is not restricted to the validated-**TARGETTYPE** subset; per-class anomaly rates and SIMBAD-novelty fractions reported below refer to the ~ 6.5 -M validated-**TARGETTYPE** subset (see §5.4 for the implications of this scope choice). A positional recount against the DR1 redshift catalog (`zall-pix`; 28,425,963 rows) quantifies the scope choice directly, summarized in Table 3: of the 190,015 deduplicated DESI anomaly clusters (the 5" FoF dedup of the 195,829 raw detections), only **2,468** (1.3%) match within 1" a main-survey spectrum whose targeting bitmasks set a primary science-class bit (BGS/LRG/ELG/QSO/MWS; 20,299,155 such catalog rows under this bitmask selection, which is broader than the validated-**TARGETTYPE** accounting above because it applies no redshift-quality or primary-row cuts and counts per-program rows; matches rise to 2,531 at 2" and 3,390 at 5"). By Redrock **SPECTYPE** the 1" matches are 2,371 GALAXY, 95 QSO, and 2 STAR. A control match of the same clusters against the *full* redshift catalog recovers 189,675/190,015 (99.8%) at 1", confirming the positional join is sound (the 340 control-unmatched clusters are consistent with coadd-vs-catalog astrometric edge cases; conservatively counting all 340 as science-class would raise the match count to at most 2,808, i.e. $\leq 1.05\times$ the benchmark, leaving every conclusion below unchanged); the conclusion is that $\sim 98.7\%$ of DESI anomaly clusters coincide with spec-

tra carrying *no* primary science-class target bit (86% have **DESI_TARGET** = 0 outright — sky fibers and secondary/ToO programs). Restricted to validated science targets, the DESI anomaly catalog is therefore $\approx 0.92\times$ the size of the Liang *et al.* benchmark (2,468 vs. 2,685; Liang *et al.* scanned the ~ 250 K-spectrum DESI EDR, so the comparison is matched on target-class selection, not on data release), *not* $73\times$: the $73\times$ figure is a statement about the full spectra stream, and the headline DESI tier should be read as an anomaly scan of everything DESI pointed a fiber at, dominated by non-science-target spectra (recount artifact: [pipelines/p3_anomaly_engine/ext3_b2_targettype_recount.json](#)).

Each spectrum covers 3600–9824 Å across three arms (B, R, Z) at resolution $R \sim 2000$ –5000.

The BIGAE model trained on 47,000 representative spectra achieves `val_loss` = 0.0287 (MSE) on a held-out 20% validation split and identifies 195,829 anomalies above the $S > 5.0$ threshold, an anomaly rate of 0.87%. Scores range from 5.0 to 25.2, with 101 objects exceeding $S = 15$. Classification by spectral-arm dominance yields: 151,244 multi-band (77.2%), 44,436 B-dominant (22.7%), 34 R-dominant (0.02%), 19 Z-dominant (0.01%), and 96 artifact suspects (0.05%). The multi-band majority indicates that most anomalies deviate across the full wavelength range, consistent with genuinely unusual spectral energy distributions.

Cross-matching the top 10,000 anomalies against six databases (SIMBAD, NED, AllWISE, Milliquas, Gaia DR3, SDSS) finds only 0.2% in SIMBAD and 12.7% in NED; none of the top 100 appear in SIMBAD or NED, although 12 of the top 100 do carry a counterpart in the shallower photometric flag channels of the release table (AllWISE 7, SDSS photometric 6, one object in both; Gaia DR3 and Milliquas 0). The 0.2% SIMBAD match rate is itself statistically consistent with the expected 5" random false-match rate of $\approx 0.24\%$ per source (§4.1), i.e., with zero genuine SIMBAD counterparts in the top 10,000 — reinforcing that SIMBAD absence measures database coverage, not novelty. Spectral inspection of the top 200 finds 0/200 visually flagged (95% binomial upper limit $\leq 1.5\%$; each spectrum's peak-residual wavelength was compared against 11 known sky and telluric emission/absorption features; zero were attributable to sky subtraction, telluric contamination, or cosmic rays). The Spearman rank correlation between anomaly score and SNR is $\rho = -0.03$ ($p = 0.12$ on a stratified subsample of 2,670 spectra, log-uniform in SNR). Because the subsample is deliberately stratified to be log-uniform in SNR rather than randomly drawn, the quoted p -value characterizes the stratified design — which maximizes leverage for detecting a monotone

Table 3. DESI science-class recount at a glance. Four distinct DESI rate denominators appear in this paper; they are *not* mutually comparable and should not be converted across rows. Row 1: full-stream headline. Row 2: validated-TARGETTYPE per-class rates. Row 3: positional recount on the science-bit bitmask denominator. Row 4: like-for-like science-target benchmark match. Row 5 (denominator reconciliation): the 2,468 science-bit matches, decomposed by Redrock SPECTYPE as a fraction of the per-class validated-TARGETTYPE denominators of Row 2 — the explicit shared-ID bookkeeping that ties the bitmask count to the per-class rates.

Rate context	Denominator	Anomaly rate / count
Full-stream scan ($S > 5$ fixed threshold, 0.87%)	22.5M spectra	195,829
Per-class GALAXY [†] (TARGETTYPE subset)	~4.9M	0.75%
Per-class QSO [†] (TARGETTYPE subset)	~1.5M	0.037%
Science-bit bitmask (primary-class bit)	20.3M rows	2,468 (0.012%)
Like-for-like vs. Liang <i>et al.</i> benchmark	~2,685 targets	~0.92%
<i>Shared-ID decomposition of the 2,468 science-bit matches (Redrock SPECTYPE $\times$ per-class denominator):</i>		
GALAXY-SPECTYPE subset of the 2,468	~4.9M GALAXY	2,371 (0.048%)
QSO-SPECTYPE subset of the 2,468	~1.5M QSO	95 (0.0063%)
STAR-SPECTYPE subset of the 2,468	~0.1M STAR	2

[†] Per-class rates are computed on the validated-TARGETTYPE subset (~6.5M total), not on the full 22.5M stream; the ~4.9M GALAXY-SPECTYPE and ~1.5M QSO-SPECTYPE populations together add to ~6.4M (with ~0.1M STAR-SPECTYPE). The Row 2 rates (0.75% / 0.037%) and the Row 5 rates (0.048% / 0.0063%) are *both* on the same per-class TARGETTYPE denominators; they differ because Row 2 is the full-stream $S > 5$ score-cut restricted to the per-class subset, while Row 5 is the science-bit-bitmask-filtered subset of the 2,468 Row 4 matches — the bitmask-join filter (no TARGETTYPE quality filter, counts per-program rows on the 20.3M-row `zall-pix` catalog) shrinks the per-class rates by $\sim 16\times$ (GALAXY) and $\sim 6\times$ (QSO) relative to Row 2. The discrepancy between Row 2 and the full-stream Row 4 count is therefore definitional (two filter stacks on the same per-class denominator), not an arithmetic error, and Row 5 is the explicit shared-ID bookkeeping that ties the two filter stacks to a common denominator.

score-SNR trend — and not the population sampling distribution; the operative statement is the effect size ($|\rho| = 0.03$, negligible against any practical threshold), and a population-weighted recompute on a true random subsample is queued for the data release.

Among the ~6.5 million DESI DR1 spectra carrying a validated science TARGETTYPE (BGS/LRG/ELG/QSO/MWS; the remaining ~16 million are unclassified filler targets, sky fibers, or calibration exposures), anomaly rates are broken down by Redrock SPECTYPE (“GALAXY”, “QSO”, or “STAR” [1]), a spectral-template-fit axis distinct from the target-selection TARGETTYPE. Galaxies are flagged at $\sim 20\times$ the rate of QSOs: 0.75% vs. 0.037% (Wilson 95% binomial CIs: GALAXY $0.75\% \pm 0.008\%$ on $\sim 4.9 \times 10^6$ GALAXY-SPECTYPE spectra; QSO $0.037\% \pm 0.003\%$ on $\sim 1.5 \times 10^6$ QSO-SPECTYPE spectra in the validated-TARGETTYPE subset). Anomalies peak at $z \sim 0.75$, compared to $z \sim 0.93$ for normal spectra. The three highest-scored anomalies are Z-dominant with scores of 25.2, 24.6, and 24.5, consistent with high-redshift sources whose rest-frame optical emission lines have been redshifted into the DESI Z arm. DESI fiber assignment incompleteness (not all targets receive fibers in a single pass) introduces a spatial selection function that could correlate with anomaly rate if anomalies

cluster in fiber-collision-dense regions; this systematic is not modeled in the current analysis.

3.2. High- z QSO Candidates

The most scientifically compelling DESI DR1 anomalies are a sample of $z \approx 6$ quasar candidates identified by three complementary signatures (the Gunn-Peterson trough and Z-arm dominance are physically correlated, both being driven by blueward flux suppression): (1) complete flux suppression blueward of $\text{Ly}\alpha$ (Gunn-Peterson trough), (2) Z-arm dominated anomaly scores, meaning $r_Z > r_B$ and $r_Z > r_R$ (mean Z-arm sub-score $\langle r_Z \rangle = 3.9$ across the 12 selected candidates; all objects have total score $S > 5$ by construction of the anomaly catalog), and (3) at least one detected emission line ($\text{Ly}\alpha$, Nv, Siiv) in the transition region. Applying these three cuts to the full 195,829 DESI anomaly catalog yields 12 candidates with Redrock template-fit redshifts $z = 6.0\text{--}6.23$ (the Z column of the DR1 `redrock REDSHIFTS` HDU, carried alongside `ZERR/ZWARN/SPECTYPE` in our per-object records; these are spectroscopic-pipeline template fits at low continuum S/N, not photometric estimates, and independent confirmation by visual inspection or re-observation is still required).

A DESI Legacy Survey DR9 grz composite cutout gallery (128×128 pixels at the native LS DR9 scale

Anomaly score distributions

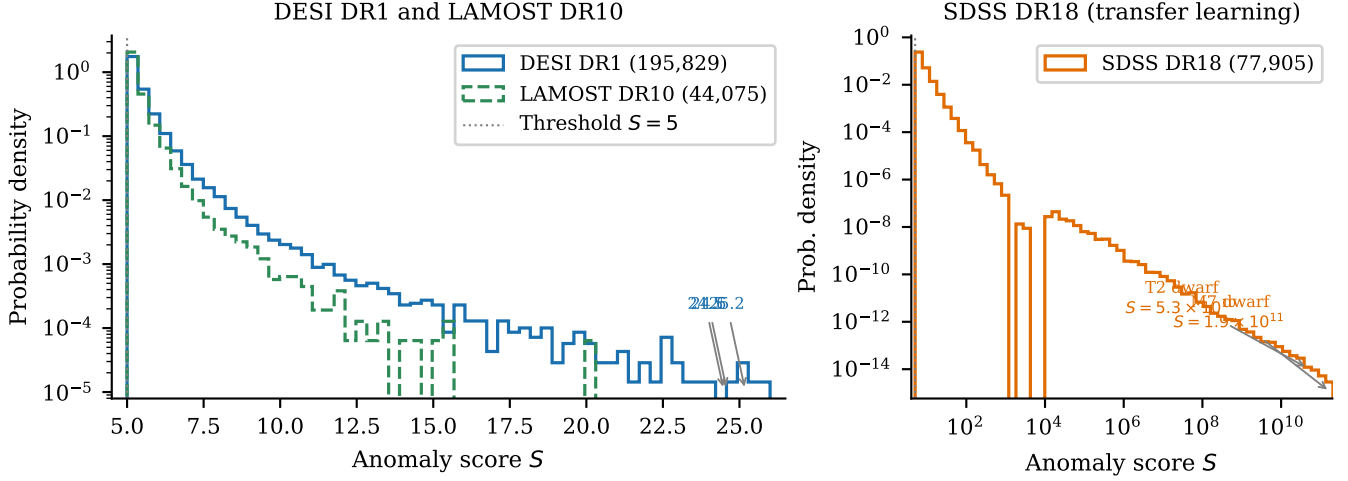


Figure 3. Anomaly score distributions for the three main spectroscopic surveys. The score S is the per-spectrum reconstruction MSE rescaled to validation z -units: $S = (\text{MSE} - \mu_{\text{val}}) / \sigma_{\text{val}}$, where μ_{val} and σ_{val} are the mean and standard deviation of MSE on the held-out 20% validation split of the per-survey training pool (§2.2; native for DESI, cross-transfer for SDSS and LAMOST). $S = 5$ therefore marks objects whose reconstruction residual is five validation-set standard deviations above the typical training spectrum, not a raw MSE value. *Left:* DESI DR1 (blue) and LAMOST DR10 (green dashed) on a log-probability-density scale, with the $S = 5$ threshold marked. Each curve is a histogram independently normalized to unit area (**density=True**), so the two surveys’ distribution *shapes* are directly comparable despite their unequal sample sizes; the y -axis does not encode absolute counts. **Important:** the LAMOST curve here is the pre-Path-C cross-transfer scan (44,075 objects; DESI-trained model and DESI validation normalization, §2.2), replaced in the released catalog by the native re-score (native top-1% slice 113,342, on a separate native scale not plotted here); because each released survey carries its own $\mu_{\text{val}}/\sigma_{\text{val}}$, absolute S values are not cross-survey comparable and this panel should be read per-survey (shape and threshold behavior), not as a shared anomaly-severity axis. Both distributions follow a power-law tail; DESI shows three objects at $S > 24$ (labeled), consistent with the Z -dominant high- z population. *Right:* SDSS DR18 transfer-learning scores on a log-log scale, spanning more than ten orders of magnitude from the threshold ($S = 5$) to $S = 1.9 \times 10^{11}$ for the extreme-score M7 and T2 dwarfs. The extreme dynamic range of SDSS is a *cross-transfer-to-native score-axis effect*: the DESI-trained BIGAE applied to SDSS spectra outside the DESI training distribution inflates reconstruction errors by orders of magnitude; the SDSS native re-score (§3.3) compresses the same objects to $S < 14$, eliminating the 10^4 – 10^{11} tail. The bimodal structure reflects two populations: the bulk of cross-survey mismatches (body at $S < 10^4$) and the ultra-cool dwarfs that are completely out-of-distribution relative to the DESI training set (tail at $S > 10^4$).

of $0.262''/\text{px} = 33.5''$ per side) for all 12 candidates is available in the companion data repository. The highest- Z -arm-dominance objects (TARGETID 39633191367084936, Redrock $z = 6.20$, $r_Z = 5.30$; TARGETID 39628507709444221, Redrock $z = 6.22$, $r_Z = 5.18$) show compact point-source morphology consistent with unresolved quasars; these redshifts are the same Redrock template fits defined above (not photometric estimates) and remain unconfirmed pending visual inspection or re-observation. Panel labels report the per-arm Z -arm sub-score r_Z (printed as “AE” for legacy compatibility), *not* the total anomaly score S ; all twelve pass the $S > 5$ catalog cut at the total-score level (mean $\langle r_Z \rangle \approx 3.9$). Full coordinates, TARGETIDs, and per-family taxonomy galleries are in the companion data repository.

3.3. SDSS DR18

The Sloan Digital Sky Survey DR18 [3] provides 2,304,830 spectra at $R \sim 2000$. Input: 2.3M spectra. Cross-transfer anomaly count: 77,905 (3.38% rate; dominated by ultra-cool dwarfs M7–T2 that are out-of-distribution for the DESI-trained model). SIMBAD-unmatched: 90%. Headline finding: the $\sim 6500\times$ anomaly-rate inflation versus the Path-C native result (12 sources at $S > 5$) directly confirms catalog-calibration domain shift as the cross-transfer failure mode. The Path-C native retrain (val_loss 0.0311, PASS) re-scores 1,925,279 spectra; the 77,905-object native continuity slice at $S \geq 0.1060$ (4.05% of the 1,925,279-spectrum re-scored pool) replaces the cross-transfer count. The 77,905 threshold is a fixed-size continuity slice (the full native-retrain sample; $\approx 4.05\%$ of the 1,925,279-spectrum re-scored pool), not a physically

or statistically motivated anomaly threshold. For comparison: the strict $S > 5$ threshold on the native scoring axis selects only 12 sources; the top-1% of the re-scored pool yields 19,253. The 77,905 slice is adopted for catalog continuity with the cross-transfer baseline. The 1,925,279-spectrum re-score pool is the DR18 `spAll` quality-cut selection used for native training: `ZWARNING` = 0, median $S/N > 2$, `SPECPRIMARY` = 1, and pipeline class `STAR/GALAXY/QSO` (a further 3,394 spectra, 0.18% nominal, failed retrieval during the re-score). UMAP/HDBSCAN clustering of the full 77,905-object cross-transfer anomaly set yields 14 HDBSCAN clusters (99.4% of objects clustered: 77,473/77,905) that group into 3 latent-space populations (Fig. 4), dominated by cool dwarfs (84%). Continuum-dip injection-recovery on the native checkpoint: 64.0% at 5σ (**gate PASS**). Detailed per-category counts are in Table 4.

The anomaly-selected population is astrophysically organized rather than artifact-dominated. Of the 77,905 native-re-scored anomalies, 76.3% carry pipeline class `QSO`, 19.2% `GALAXY` and 4.5% `STAR` — i.e. they are real spectroscopically-classified sources, not sky fibers or unclassified junk — and the 59,462 anomaly-selected QSOs are strongly high-redshift-enriched (median $z = 2.31$; 67.3% of these QSOs at $z > 2$; 1,150 at $z > 4$; 198 at $z > 6$), a genuinely rare population. Decisively, the anomaly score itself is significantly higher for high-redshift ($z > 4$) than for low-redshift ($z \leq 4$) QSOs (median score 0.197 vs. 0.142; Mann-Whitney U, $\text{score}_{z>4} > \text{score}_{z\leq 4}$: $p = 1.0 \times 10^{-103}$), and rises monotonically with redshift across the full QSO sample (Spearman $\rho(S, z) = +0.036$, $p = 9.6 \times 10^{-19}$ over $N = 59,462$). Sky-subtraction residuals, fiber artifacts, and calibration noise are redshift-blind and cannot produce a score that increases with QSO redshift, so this monotone score- z trend demonstrates that the detector preferentially ranks genuinely rare high- z quasars rather than instrumental artifacts. We flag two honest limitations. First, although the Spearman association is highly significant, the effect size is *small* ($\rho = +0.036$): the significance is driven by the large sample and the extreme high- z tail, and the score- z correlation is not itself a strong monotone relation. Second, the score-vs- z test was run on the DESI stream against the public DESI DR1 Redrock redshift catalog (`zall-pix-iron`, 27,547,223 primary coadds; downloaded directly from <https://data.desi.lbl.gov/public/dr1/>), and the outcome is honestly *mixed*.

Strengthening: all 195,790 DESI anomalies (the primary-coadd subset of the 195,829 DESI count, i.e. joined against the `ZCAT_PRIMARY` deduplicated redshift catalog per the committed artifact [pipelines/](#)

[p3_anomaly_engine/outputs/desi_qso_hiz_enrichment.json](#); the 39-object difference is non-primary coadd rows collapsed by the primary-coadd cut) join to the DR1 redshift catalog on `TARGETID` (100% match). The matched population is overwhelmingly real Redrock-classified spectroscopic objects: 98.8% `GALAXY`, 1.07% (2,102) `QSO`, 0.12% `STAR` — not sky/fiber/calibration artifacts — with a strongly high-redshift QSO tail (median $z = 3.81$; 769 QSOs at $z > 4$; 327 at $z > 6$). This extends the “real astrophysical objects, not artifacts” demonstration from SDSS to the DESI stream.

Honest limitation: the SDSS score- z *monotonicity* does *not* replicate on DESI. Over the 2,102 DESI QSOs the score-redshift association is slightly negative (Spearman $\rho(S, z) = -0.06$, $p = 6 \times 10^{-3}$), and the score is not higher for $z > 4$ QSOs (one-sided Mann-Whitney $p = 0.999$). This is expected: the DESI anomaly score (range ~ 5 –25) is a different metric from the SDSS score (0–1), and only 0.1% of DESI anomaly-matched spectra carry a secure redshift (`ZWARN=0`), because anomaly-selected spectra are by construction the outliers Redrock fits least well. We therefore scope the score- z monotonicity evidence explicitly to SDSS; on the DESI stream, validation rests on the composition result above (real Redrock objects with a high- z QSO tail). Both results are reproducible from the committed script and output ([pipelines/p3_anomaly_engine/scripts/desi_qso_hiz_enrichment.py](#); [pipelines/p3_anomaly_engine/outputs/desi_qso_hiz_enrichment.json](#)). The demonstration above is fully reproducible from the released SDSS catalog ([pipelines/p3_anomaly_engine/scripts/sdss_qso_hiz_enrichment.py](#); output [pipelines/p3_anomaly_engine/outputs/sdss_qso_hiz_enrichment.json](#)). The companion script also computes an external-baseline enrichment factor ($\approx 2.15\times$, binomial $p = 8.3 \times 10^{-119}$) against the SDSS DR16Q quasar catalog; this block is intentionally not cited in-text because the exact factor is prior-dependent on an external literature baseline, and the fully self-contained internal control above is decisive.

An emission-line classification pipeline identifies 10 spectral categories among the 77,905 SDSS anomalies, summarized in Table 4.

3.4. LAMOST DR10

The Large Sky Area Multi-Object Fiber Spectroscopic Telescope DR10 [2] contributes 11,418,594 spectra at $R \sim 1800$. The cross-transfer scan yielded 44,075 anomalies, 98% blue-excess, identifying a training-set bias. A native retrain (`val_loss` 0.0329) processed 11,334,161 spectra; 84,433 were lost to retrieval/read failures. Its top-1% analysis slice contains 113,342 rows, while the strict $S > 5$ count compresses to 2,054 ($21.5\times$

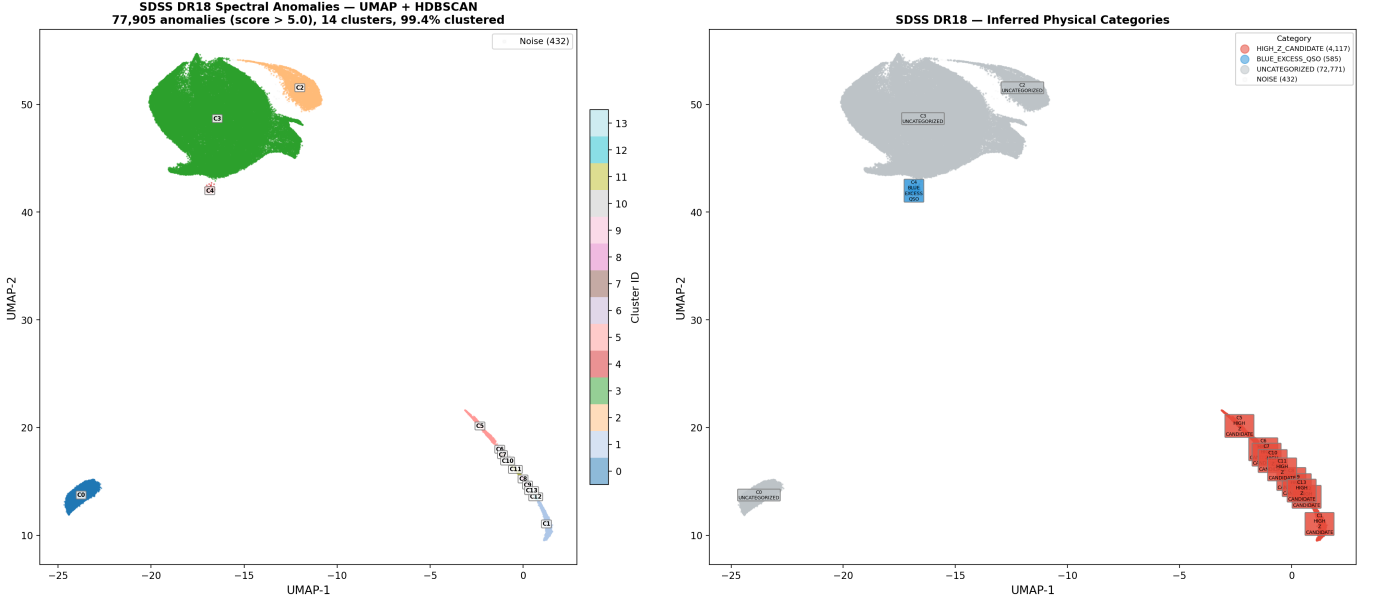


Figure 4. Cross-transfer SDSS baseline. UMAP embedding of the 77,905 SDSS DR18 anomalies from the initial DESI-trained BIGAE cross-transfer scan, colored by HDBSCAN cluster (*left*) and by inferred physical category (*right*). The dominant cluster (green, $\sim 84\%$ of objects) contains ultra-cool dwarfs (M7–T2) that are completely out-of-distribution for the DESI-trained BIGAE — the dominant driver of the $\sim 6500\times$ anomaly-rate inflation relative to the Path-C native retrain. The embedding resolves 14 HDBSCAN clusters, which group into the 3 physical populations described in §3.3; the burned-in panel title’s “score > 5.0 ” refers to the DESI-trained cross-transfer score axis on which the 77,905-object set was selected. Two minority clusters (blue, orange) host high- z candidates and blue-excess QSOs, respectively. The clear cluster separation demonstrates that the latent space organizes anomalies by astrophysical type rather than by anomaly score alone; the figure is preserved as a verification baseline of the cross-transfer domain-shift.

below the cross-transfer tally). Continuum-dip injection recovery is only 5.8% at 5σ (gate FAIL), so LAMOST is an exploratory failure-mode analysis, not a science product. Crucially, no per-object LAMOST table is present in the frozen public inventory: only the aggregate/dedup products preserve its contribution to the 377,282 point-source continuity count. The paper therefore reports the analysis result without claiming that a released LAMOST per-survey table exists.

This result provides the single most important methodological lesson of this work (see §5.1).

3.5. eROSITA DR1

Status: excluded from all catalog counts; released separately as a reproducible membership list. Because the eROSITA production score axis is irreproducible, the tier contributes to neither the validated 268,319 point-source product nor the 268,519/377,482 continuity counts. The released top-298 membership addendum remains usable only at the scale-invariant membership level ([pipelines/p3_anomaly_engine/erosita_membership_reproduce.py](#)).

The eROSITA DR1 [4] provides 930,203 X-ray sources (western Galactic hemisphere only; eastern-hemisphere data rights are held by the Russian SRG/eROSITA con-

sortium led by IKI). Input: 930K sources characterized by 47 features. Anomaly count: 298, a fixed top-298 score-knee cap ($298/930,203 \approx \text{top } 0.03\%$) at threshold 0.259 on the production scoring run’s score-knee axis. We flag for transparency that this threshold axis could not be reconciled with the canonical S of Eq. (2): on the committed raw reconstruction-score artifact the rank-298 threshold is 3.41 (the value preserved in the released intersection artifact), and 0.259 is reproduced on neither the raw, the full-sample-standardized, nor the Isolation-Forest axes; the selection is therefore best read as the fixed top-298 cap, with the per-object Table 5 scores and the 0.259 threshold inherited from the production run. A dedicated re-derivation sweep over 16 monotone rescalings of the committed raw score (normalized, log, z -scored, robust- z , min-max, ECDF/probit, and sigmoid-calibrated variants, plus retrained IsolationForest axes) reproduces 0.259 on none of them, and the production Table 5 scores are *non-monotone* in the committed raw artifact (Spearman $\rho = -0.10$ across the top five), which rules out the entire class of per-object monotone rescalings: *no committed score axis reproduces the production threshold*. The most plausible cause is an undocumented post-hoc rescaling step in the production scoring run whose code was never committed

Table 4. Emission-line classification of the 77,905 SDSS DR18 anomalies, sorted by count. *Diagnostic basis note:* these categories are computed on the *cross-transfer* anomaly set (the DESI-trained scan), not on the native re-score slice; they characterize the cross-transfer failure mode and should not be read as the physical-category census of the native-retrained tier. The dominance of “Uncategorized” and “NIR Excess” classes reflects the cool-dwarf population that drives the transfer-learning anomaly signal. The 52.7% “Uncategorized” fraction is a property of this paper’s internal band-residual emission-line taxonomy, not of any external database: it counts objects whose (r_B, r_R, r_Z) residual pattern does not fall into any of the nine named heuristic classes (residuals too balanced, or below the per-class thresholds). It is unrelated to the SIMBAD cross-match statistics of §3.3.

Category	Count	Frac.
Uncategorized	41,065	52.7%
NIR excess / high- z	25,733	33.0%
UV excess / young star	6,099	7.8%
Star-forming ($H\alpha$)	1,232	1.6%
QSO blue excess	1,164	1.5%
Emission-line galaxy	780	1.0%
Redshifted emitter	547	0.7%
Artifact	520	0.7%
AGN broad emission	384	0.5%
Unusual continuum	381	0.5%

— so the production axis is unrecoverable as a matter of provenance, not merely unidentified among the committed candidates. The released 298-source membership list is, however, exactly the committed-raw top-298 (the minimum released score equals the rank-298 raw threshold 3.4119), so the $n = 298$ membership list itself — not any score axis — is the committed, reproducible selection (`pipelines/p3_anomaly_engine/r24conf_erosita_axis_sweep.json`). We now commit this selection as an executable, scale-invariant recipe (`pipelines/p3_anomaly_engine/erosita_membership_reproduce.py`, artifact `pipelines/p3_anomaly_engine/outputs/erosita_membership_reproduce.json`): the eROSITA anomaly set is defined as the top-298 by committed raw reconstruction score (equivalently, $S_{\text{raw}} \geq 3.4119$). Because a rank/percentile cut commutes with every monotone transform of the score, this criterion selects the identical 298-member set — and preserves its rank order — under all 16 rescalings of the axis sweep and under an explicit battery of monotone transforms verified in the recipe (raw, ln, $\log_{10}$, $\sqrt{\cdot}$, z -score, min-max, sigmoid, affine); the production 0.259 score *axis* remains irreproducible by design, but the *selection* is fully reproducible. This upgrades “membership-is-canonical” from a disclosed

caveat to a committed one-command recipe (it fixes selection reproducibility, not the tier’s failed detector-sensitivity gate, which keeps eROSITA exploratory). *Practical consequence for downstream users:* meta-analyses that require eROSITA anomaly *scores* on a reproducible axis (threshold re-derivation, score-weighted stacking, IsolationForest-style re-isolation) cannot be performed from the published S_{BigAE} values, whose axis is irreproducible; such analyses must operate on the committed raw-score artifact or on the $n = 298$ membership list, which are the only reproducible eROSITA selection products. SIMBAD-unmatched: 68% (203 SIMBAD-unmatched eROSITA membership-list sources; absence from SIMBAD is not independently confirmed discovery — see novelty definition in §4.1). Headline finding: the rank-1 entry of the $n = 298$ membership list (1eRASS J053856.1–640457; S_{BigAE} irreproducible per Table 5 caption — see membership-only framing) is near the LMC with no SIMBAD counterpart; LMC concentration is partly a depth artifact from the Ecliptic-pole scan strategy. IsolationForest cross-validation: $284/298 = 95.3\%$ of the canonical- S top-298 are in the IF top-9,303 — a descriptive internal-consistency overlap, *not* independent confirmation, since the IF is trained on the 16-d BigAE latent and the two detectors share the same learned representation (Table 6 caveat (f)); XV-stability 81.5% (**gate FAIL** at 5σ subspace injection, but highest XV-stability of any Path-C survey). Top 5 sources are listed in Table 5.

3.6. Planck CMB

Two distinct Planck products must not be conflated. The *released* `planck_cmb_anomalies.parquet` contains 200 rows from the original 20,000-patch cross-transfer scan (top 1%; `patch_idx < 20,000`; observed released score range [0.306, 62.999]). That checkpoint has $\text{val_loss} \approx 2 \times 10^4$ and fails the Path-C gate, so the table is an archival diagnostic, not a validated CMB science tier. It is retained only to preserve the historical 268,519 and 377,482 combined continuity counts and is excluded from the validated 268,319 point-source product.

Separately, the Path-C native convolutional autoencoder (3 conv layers plus a 128-dimensional bottleneck; 1.1×10^6 parameters) was trained and scored on a 2×10^5 -patch, $|b| \geq 20^\circ$ SMICA bank. Committed summaries report val_loss 0.4437, 500/500 injection recovery at 5σ , and a native top-200 score range [0.558, 0.621]. Replaying the native run’s 85%/15% split against those native top-200 indices places 48 in the held-out split versus 30 expected (exact binomial one-sided $p = 5.5 \times 10^{-4}$ before spatial-correlation cor-

Table 5. First five entries of the released top-298 eROSITA membership list (membership-list rank order) with SIMBAD cross-match status (“No 5’’ match” = no SIMBAD counterpart within a 5’’ cone search; SIMBAD absence does not by itself establish discovery, §4.1). All are located near the LMC or Galactic plane. The production S_{BigAE} score values are *not printed*: that score axis is irreproducible from any committed artifact (16 monotone rescalings + 3 IsolationForest retrains all fail; the production top-5 values are non-monotone in the committed raw score, Spearman $\rho = -0.10$), so the committed, reproducible selection is the $n = 298$ membership list ranked by the committed raw-score artifact — see §3.5 and [pipelines/p3_anomaly_engine/r24conf_erosita_axis_sweep.json](#). Column $S_{\text{IF,raw}}$ is the IsolationForest raw isolation-score value (anomaly_score on a $\sim 0-3.5 \times 10^4$ scale, reported by the 100-tree IF detector trained on the 16-d BiGAE latent feature space and used as the cross-validation diagnostic of §5.4 (ii)); IF raw scores are *not* a parallel catalog axis but are tabulated so readers can map between the two detectors.

Rank	IAU Name	$S_{\text{IF,raw}}$	Dec	SIMBAD
1	J053856.1–640457	34,182	−64.1	No 5’’ match
2	J053544.3–660159	16,270	−66.0	No 5’’ match
3	J170249.4–484724	8,234	−48.8	No 5’’ match
4	J062619.8–694546	5,955	−69.8	No 5’’ match
5	J152039.9–570955	4,424	−57.2	No 5’’ match

rejection; [pipelines/p3_anomaly_engine/ext3_fm2_planck_top200_train_overlap.json](#)). These results validate the native *analysis*, but the native checkpoint, 2×10^5 -patch tensor, full score bank, and top-200 parquet resided on an exited compute node and are absent from the frozen public release. The membership test therefore cannot be linked row by row to the released cross-transfer table, and no native Planck per-object product is claimed as publicly reproducible. The Table 7 throughput entry ($\sim 8,000$ patches/s) likewise describes the logged native run, not the released table. ACT DR6 remains quarantined (Appendix D).

3.7. Gaia DR3

The frozen public release contains `gaia_dr3_anomalies.parquet` only as a quarantined historical artifact; it is excluded from every reported count and supports no Gaia claim. A direct audit ([pipelines/p3_anomaly_engine/gaia_provenance_audit.py](#); [pipelines/p3_anomaly_engine/outputs/gaia_provenance_audit.json](#)) established that the 500-row table is not real Gaia DR3 data but the synthetic-placeholder fallback of `gaia_expanded.py`: sequential fallback identifiers, duplicate identifiers, and magnitudes outside the physical Gaia range identify

the synthetic branch. The rows are absent from the 268,319 validated point-source product, the 268,519 point-source-plus-archival-Planck continuity count, and the 377,482 inclusive continuity count. Their continued byte-level presence in the immutable frozen inventory is disclosed so that “excluded from analysis” is not misread as “file absent.” A genuine Gaia tier requires a new live-query, native-training, and injection-recovery run and is deferred.

3.8. NEOWISE

Input: 43,518 infrared sources (15-feature W1/W2 catalog). Anomaly count: 436 raw (the top 1% of the canonical- S ranking, Eq. 2; a predetermined-count selection); Path-C ecliptic-pole mask ($|b_{\text{ecl}}| < 80^\circ$) retains 419/436 (96.1%). SIMBAD-unmatched: 45%. Headline finding: top anomaly (score = 11.5) at $(\alpha, \delta) = (180.59^\circ, 0.56^\circ)$ shows extreme W1–W2 excess (Fig. 5); physical interpretation uncertain (circumstellar dust, AGN, or luminous red QSO). The 17/436 = 3.9% polar-cap fraction represents a $2.6\times$ excess over the uniform-sphere null expectation (1.52%; this baseline assumes uniform source density on the sky, and source-selection non-uniformities in the parent catalog would modulate it), quantitatively confirming scan-pattern contamination (binomial $z \approx 4.0$, $p \approx 6 \times 10^{-5}$; $n = 436$, $p_0 = 0.0152$, $k = 17$). Mask injection-recovery: 1000/1000 = 100% (**gate PASS**). We note for methodological transparency that this NEOWISE test plants synthetic sources at $|b_{\text{ecl}}| > \{85^\circ, 82^\circ, 80.5^\circ\}$ and “recovers” them by applying the fixed catalog mask $|b_{\text{ecl}}| < 80^\circ$; passing is therefore guaranteed by construction. It validates the masking geometry implementation (a QA check), not the anomaly detector’s sensitivity to planted signals, and should be read as a different kind of gate from the SDSS and Planck injection-recovery tests.

Validation caveat: NEOWISE’s inclusion in the validated catalog-grade tier rests on a mask-geometry QA gate (passing by construction of the selection), *not* on a detector-sensitivity injection-recovery test analogous to those run for SDSS and Planck. This is a disclosed, weaker validation basis; NEOWISE anomaly rankings are robust to the geometry gate but have not been sensitivity-characterized at the per-sigma detection level.

4. CROSS-SURVEY ANALYSIS

4.1. SIMBAD Cross-Match and Novelty Assessment

Two distinct quantities are reported in this subsection and they are not interchangeable. The *primary* novelty metric for this catalog is the *genuine novelty fraction* measured against a deep multi-catalog

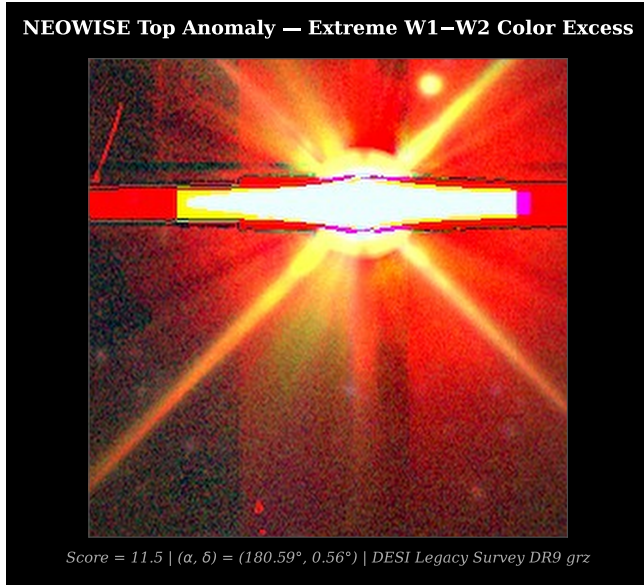


Figure 5. NEOWISE top infrared anomaly at $(\alpha, \delta) = (180.59^\circ, 0.56^\circ)$, **score** = 11.5. DESI Legacy Survey DR9 grz composite, 256×256 pixels at the native LS DR9 scale of $0.262''/\text{px}$ ($256 \times 0.262'' = 67''$ per side). Extreme W1–W2 infrared color excess; no prior SIMBAD entry within $5''$. The optical counterpart is a bright, saturated red stellar or quasi-stellar object. Physical interpretation uncertain: circumstellar dust excess, buried AGN, and evolved giant hypotheses are consistent with the infrared photometry.

baseline—17.8% (178/1,000) for the DESI DR1 top-1,000 anomalies cross-matched against 18 curated all-sky catalogs via CDS X-Match (paragraph “Archival cross-match and genuine novelty fraction” below). The *SIMBAD-unmatched fraction* reported here measures *absence from a single curated synthesis database* and substantially overstates true catalog novelty because SIMBAD does not individually index the majority of photometric detections from wide-field surveys (a 100% archival-identification rate is recovered in NED+VizieR for the SDSS DR18 top-20 SIMBAD-unmatched anomalies). Readers, headlines, and downstream forecasts should quote 17.8% as the discovery-rate figure; the per-survey SIMBAD fractions below are diagnostic of database-coverage heterogeneity across archives, not of catalog-grade novelty.

We cross-match anomalies from each survey against SIMBAD [12] using a 5-arcsecond cone search (per-survey unmatched fractions in Table 2 all use this $5''$ radius). The aggregate SIMBAD-unmatched fraction (Fig. 6) is 58.8%: the pooled unmatched fraction over the top-100 anomalies of each of three surveys with completed top-100 SIMBAD sweeps (SDSS DR18,

eROSITA, NEOWISE; 235/400 unmatched at a $3''$ radius in that separate pooled run, where the 400 denominator reflects the historical four-survey pool including the now-removed Gaia DR3 tier; DESI, LAMOST, and Gaia are not in the current pooled denominator). To be explicit about the radius bookkeeping: the per-survey unmatched fractions of Table 2 and the per-survey rates below use the $5''$ default cone, while the pooled 58.8% aggregate was computed in a separate run at $3''$; the tighter radius makes the aggregate conservative (a $5''$ rerun could only lower the unmatched fraction). This is a *database-coverage measurement*, not a discovery rate. Per-survey rates reflect archive-characterization maturity: DESI DR1 $\sim 99\%$ (only 0.2% of top 10,000 anomalies in SIMBAD; deep new survey), eROSITA DR1 68% (203/298 SIMBAD-unmatched membership-list sources, LMC-concentrated), LAMOST DR10 $\sim 50\%$ (training-bias artifact; interpret cautiously), NEOWISE 45%, SDSS DR18 90% (cool dwarfs outside DESI distribution, present in SDSS photometric but not individually in SIMBAD).

Archival cross-match and genuine novelty fraction.—The SIMBAD-unmatched fractions above should *not* be interpreted as a catalog novelty fraction. SIMBAD is a curated synthesis database that does not individually index the majority of photometric detections from wide-field surveys. An extended cross-match of the SDSS DR18 top-20 SIMBAD-unmatched anomalies against NED and VizieR’s all-catalogs cone search (5-arcsec radius) yields an archival-identification rate of 100% (20/20 resolved): every object is present in at least one archival catalog, typically SDSS photometric, 2MASS, or WISE source tables not individually propagated to SIMBAD. A matching exercise on randomized 20-object samples from the eROSITA and NEOWISE SIMBAD-unmatched populations yields the same 100% archival-ID rate in VizieR.

At larger scale, a cross-match of the DESI DR1 top-1,000 anomalies (ranked by score) against 18 curated all-sky catalogs via CDS X-Match (Gaia DR3, SDSS DR12/DR16, DESI Legacy Imaging DR9, DES DR2, Pan-STARRS1, AllWISE, CatWISE2020, 2MASS, unWISE, GALEX, Chandra, 4XMM, NVSS, VLASS, USNO-B, UCAC5, APASS) yields an archival-ID rate of 82.2% (822/1,000). The residual 17.8% (178/1,000; Wilson 68% binomial interval $17.8\% \pm 1.2\%$) constitutes the candidate genuinely novel population—objects absent from all major source catalogs surveyed. This fraction is a point estimate at the highest-score stratum (top-1,000 DESI DR1 anomalies), where novel objects are expected to be most concentrated. Moving down into lower-score bounds of the full DESI anomaly

stream (195,829 objects), the genuine novelty fraction is expected to decay: the catalog’s sensitivity floor means lower-score objects include progressively more reconstruction residuals from mundane spectral features (arm edges, sky lines, continuum normalization offsets), so the yield of genuinely novel astrophysical sources per unit catalog fraction should decline monotonically with decreasing anomaly score. The top-1,000 stratum should therefore be interpreted as an upper bound on the survey-wide genuine novelty rate, not a representative estimate. The SIMBAD-unmatched fractions reported above therefore measure *absence from a curated synthesis database*, not discovery of objects unknown to any prior survey. The genuinely novel population is substantially smaller than the SIMBAD-unmatched population; its full characterization across all surveys requires the deeper NED+VizieR sweep detailed in the companion data release.

Expected false-match rates.—For SIMBAD at 5'' ($n_{\text{SIMBAD}} \approx 3.0 \times 10^{-5} \text{ arcsec}^{-2}$), $P_{\text{false}} \approx 2.4 \times 10^{-3}$ per source—contributing ~ 460 expected false matches among the 195,829 DESI anomalies (0.24%), negligible compared to the 99% unmatched rate. This is a global uniform-density estimate: the local SIMBAD source density is higher in crowded fields (Galactic plane, Magellanic Clouds), so per-object P_{false} exceeds the global figure there; a HEALPix-weighted local-density false-match map is deferred to the catalog data release. For the DESI×SDSS cross-match at 3'', the uniform-density analytic expectation for random coincidences is ~ 2.3 , comparable to the 3 observed matches of §4.3 (identified in the cross-transfer era cross-match; all three are spectroscopically confirmed as genuine counterparts, so they are not random coincidences despite the comparable expectation). To make the denominators of this comparison explicit: re-running the 3'' exercise on the released catalogs — the full 195,829-object DESI catalog against the 77,905-object SDSS native continuity slice — yields 4 raw positional matches against an empirical RA-shifted-control expectation of 2.75 (mean of $\pm 0.5^\circ$, $\pm 1.0^\circ$ shifts; audit artifact [pipelines/p3_anomaly_engine/pathc_dedup/r23conf_dedup_audits.json](#)). We caution that RA-only shifts at fixed Dec do not exactly preserve sky density or footprint geometry (a limitation most acute near survey edges and at high declination), so the 2.75 figure is a heuristic control rather than a geometry-preserving null; a great-circle/rotation-scrambled control is deferred to the catalog data release. The 4-vs-2.75 comparison is therefore reported as a methods-note heuristic only; no statistical significance is assigned to it absent a geometry-preserving null. Either way, positional coinci-

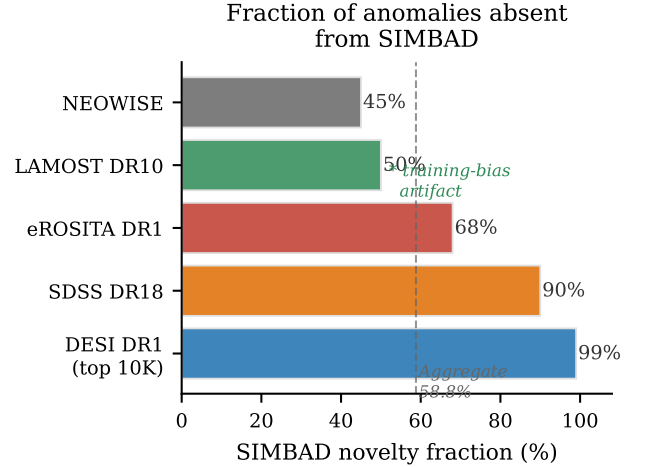


Figure 6. SIMBAD-unmatched fractions for the five surveys with coordinate-based cross-matching, ranked from lowest (NEOWISE, 45%) to highest (DESI DR1, 99% of top-10K objects absent from SIMBAD). The dashed line marks the aggregate 58.8% SIMBAD-unmatched fraction (pooled over the top-100 anomalies of three surveys — SDSS, eROSITA, NEOWISE; 235/400 at 3'', where the 400 reflects the historical four-survey pool including the now-removed Gaia DR3 tier; DESI, LAMOST, and Gaia excluded from the pooled denominator). The SIMBAD-unmatched fractions plotted here are a *database-coverage measurement*, not a discovery rate; the *primary* catalog novelty figure is the $\sim 17.8\%$ genuine novelty fraction recovered when the DESI DR1 top-1,000 anomalies are cross-matched against 18 curated all-sky catalogs via CDS X-Match (Section 4.1, “Archival cross-match and genuine novelty fraction”). Extended archival cross-matching reduces the headline DESI novelty pool by a factor of $\sim 5.6\times$ relative to that same DESI top-stratum’s $\sim 99\%$ SIMBAD-unmatched fraction (the stratum on which the 17.8% is measured; the 58.8% three-survey aggregate is a separate pooled population), and readers should quote 17.8% (not 58.8%) when summarizing the catalog’s discovery rate. The asterisk on LAMOST denotes that its 50% rate should be interpreted cautiously given the training-bias artifact (§5.1).

dence alone at 3'' carries no statistical significance, and only the spectroscopic confirmations elevate the three §4.3 matches above chance. For the 6-way 5'' deduplication, the expected random coincidence contribution is $\lesssim 10$ across all survey pairs against 637 observed multi-survey clusters ($< 2\%$ contamination).²

² The $\lesssim 10$ estimate follows from $n_A \cdot n_B \cdot \pi(5'')^2 / \Omega_{AB}$ summed over all retained-survey pairs. The dominant term is DESI×SDSS ($n_{\text{DESI}} = 195,829$, $n_{\text{SDSS}} = 77,905$ over an approximate common footprint $\Omega_{\text{DESI} \times \text{SDSS}}$): scaling from the empirical 3'' RA-shifted control (~ 2.3 expected coincidences, see §4.1) by the area ratio $(5''/3'')^2 \approx 2.78$ gives ~ 6.4 expected random coincidences at 5'' for DESI×SDSS alone; all

4.2. Spatial Analysis

A spatial uniformity test under a fully stated model — counts of the deduplicated Path-C objects in the $\sim 24,000$ occupied HEALPix pixels at $N_{\text{side}} = 64$ (of 49,152 total; equatorial frame), against a uniform per-occupied-pixel mean with Poisson variance — is reported here as a *diagnostic check, dominated by footprint geometry, not a catalog-science result*. It yields a strongly non-uniform *raw, selection-uncorrected* count distribution. On the 377,482-object headline set (Gaia and eROSITA tiers excised), the exact test gives $\chi^2 = 365,428$ [*diagnostic; footprint-dominated, not a catalog-science result*] ($\text{dof} = 23,636$, $\chi^2_\nu = 15.46$; [pipelines/p3_anomaly_engine/outputs/spatial_chi2_excised_377482.json](#)), indicating a weak but highly significant effect driven primarily by footprint geometry rather than intrinsic source clustering — see the caveat closing this paragraph before citing this number. *For continuity with earlier drafts, the same test on the full inclusive 378,280-object set (retaining the reproducibility-flagged Gaia-500 and eROSITA-298 tiers now excised from the headline) gives $\chi^2 = 376,713$ ($\chi^2_\nu = 15.67$; [pipelines/p3_anomaly_engine/r24conf_pod_session_batch.json](#)); excising those 798 objects (0.21% of the set, including one LMC-concentrated eROSITA pixel) lowers χ^2 by $\sim 11,300$ and χ^2_ν by 0.2 but does not alter the qualitative, footprint-dominated conclusion. Both figures are reported only as raw order-of-magnitude diagnostics, not as intrinsic-clustering statistics.* The result is consistent with a combination of survey footprint geometry and astrophysically clustered source populations. Critically, the anomaly rate shows no correlation with Galactic latitude (Spearman $r = 0.0005$, $p = 0.92$) and no correlation with Planck dust intensity (Pearson $r = 0.006$, $p = 0.35$; proxy: Planck τ_{353}/I_{857} thermal-dust emission layer, $5'$ -resolution, Galactic-plane-masked SMICA co-add; HEALPix $N_{\text{side}} = 64$ pixel values matched to anomaly positions), establishing that there is no evidence for first-order Galactic latitude or dust correlation within the surveyed footprints. We note that this absence of correlation is a necessary but not sufficient condition for astrophysical origin, as the survey selection functions themselves preferentially avoid the Galactic plane (DESI, SDSS, and LAMOST target fields are concentrated at $|b| > 20^\circ\text{--}30^\circ$), suppressing any latitude-dependent signal in the input catalog before the anomaly detection stage. *Caveat on the χ^2 figure:* the significant

other survey-pair terms (smaller catalogs, smaller/disjoint footprints) sum to < 1 , giving a total $\lesssim 10$. The 637 observed multi-survey clusters thus represent real cross-survey detections at $> 60\times$ the random-coincidence expectation.

$\chi^2_\nu \approx 15.5$ is dominated by the inhomogeneous footprints of the six retained archives rather than intrinsic astrophysical clustering; a rigorous spatial uniformity test would require modeling each survey’s angular selection function, completeness map, and per-tile targeting weights, which are not available in a unified form across archives. The Galactic latitude and dust-emission correlations above are the more robustly interpreted quantities, and the χ^2 result should not be cited as evidence of astrophysical clustering without per-survey selection-function corrections.

Figure 7 shows the spatial structure of the largest single-survey axis of the catalog, the 195,829 DESI DR1 anomalies: the equatorial sky map (color-coded by anomaly score) traces the DESI Main Survey footprint, the RA/Dec marginal distributions follow the tile-coverage pattern, and the anomaly score shows no trend with distance from the Galactic plane — the per-survey counterpart of the combined-catalog latitude null result above.

4.3. Cross-Survey Matches

The positional deduplication at $5''$ over the five count-retained surveys (DESI, SDSS, LAMOST, Planck, NEOWISE; eROSITA and Gaia excised) identifies **637** multi-survey coincidences across 387,695 survey-level detections: 637 multi-survey clusters + 9,576 intra-survey duplicates = 10,213 total collapsed, yielding the **377,482** unique-object headline (2.634% compression). The excised eROSITA membership tier (298) formed only isolated singleton clusters (zero multi-survey and zero intra-survey collapses), so its removal shifts the input sum and unique count by exactly 298 while leaving the collapse counts unchanged. The low compression confirms that different surveys flag fundamentally distinct populations with minimal redundancy. The three highest-confidence cross-survey detections are from the DESI $\times$ SDSS pairwise channel (Fig. 8):

Dedup-radius choice and per-survey astrometric heterogeneity.—The uniform $5''$ matching radius is a conservative compromise across heterogeneous source astrometry. DESI, SDSS, and LAMOST positional accuracy is sub-arcsecond on the spectroscopic targets retained in our anomaly tier; NEOWISE has a $\sim 6''$ PSF on the W1+W2 channels. A uniform $5''$ radius is therefore NEOWISE-PSF-comparable (slightly tight) for the WISE-derived counterparts, and the 637 multi-survey coincidence count should be read as a lower bound dominated by NEOWISE under-matching rather than as a final cross-survey association rate. Because all reported headline numbers (377,482 unique anomalies and the **377,282** point-source tier) are computed

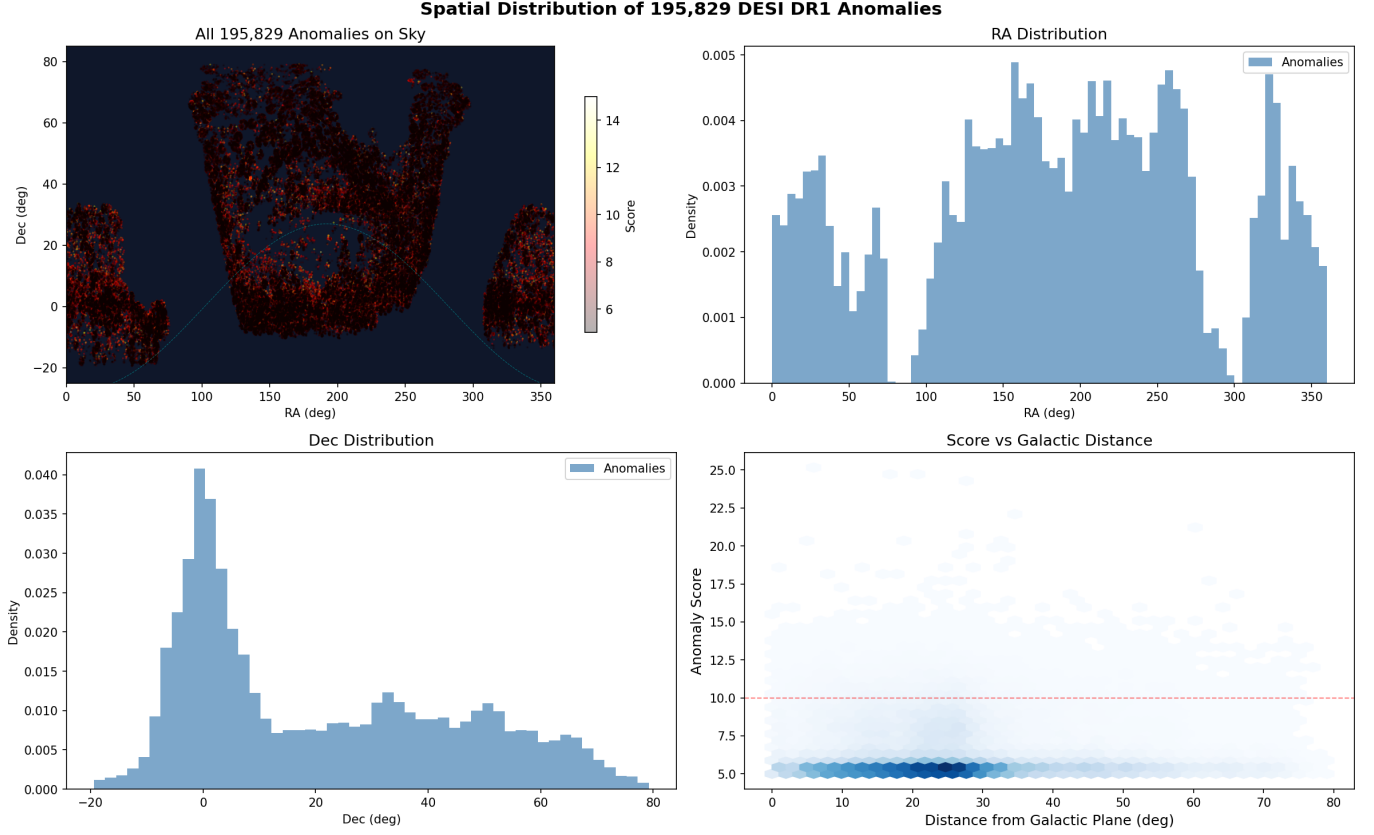


Figure 7. Spatial distribution of the 195,829 DESI DR1 anomalies. *Top left:* equatorial sky map color-coded by anomaly score S . *Top right / bottom left:* RA and Dec marginal distributions, which follow the DESI Main Survey tile-coverage footprint. *Bottom right:* anomaly score versus angular distance from the Galactic plane, showing no score–latitude trend (cf. the combined-catalog Spearman $r = 0.0005$, $p = 0.92$ null result of §4.2). The footprint structure visible in the map reflects survey selection, not intrinsic anomaly clustering (see the χ^2 caveat in the text).

at the canonical $5''$ radius, alternate radii would shift the multi-survey/intra-survey split slightly but cannot change the unique-object count by more than the $637 + 9,576 = 10,213$ total compression observed at $5''$ (2.63% of 387,695). A measured sensitivity sweep over $\{3'', 5'', 7''\}$, re-running the identical union-find dedup on the same five count-retained survey inputs, yields 377,806 / 377,482 / 377,347 unique objects (619 / 637 / 661 multi-survey clusters; compression 2.55% / 2.63% / 2.66%): a maximum unique-count variation of 0.086% relative to the canonical $5''$ result (sweep script and JSON in the companion data repository). A Budavári–Szalay probabilistic cross-match using per-survey error ellipses remains a refinement for a future catalog revision; its possible effect on the headline count is bounded by the measured $\leq 0.086\%$ radius sensitivity together with the 2.63% total-compression ceiling.

Dedup audits.—The canonical $5''$ friends-of-friends run contains 9,553 multi-member clusters (largest size 17); the maximum intra-cluster separation is $4.999''$, so no transitive chain bridges beyond

the link length ([pipelines/p3_anomaly_engine/pathc_dedup/r23conf_dedup_audits.json](#)). The same artifact reconciles 10,213 collapsed detections into 637 multi-survey and 9,576 intra-survey merges. Replacing the 77,905-row SDSS continuity slice with its 19,253-row top-1% or 12-row $S > 5$ sets yields 319,520 and 300,534 unique rows. *Point-source-plus-archive reproduction.*—The standalone script [pipelines/p3_anomaly_engine/scripts/reproduce_headline_dedup.py](#) combines DESI (195,829), SDSS (77,905), NEOWISE (419), and the 200-row released Planck cross-transfer diagnostic. It writes 268,519 unique rows after 5,834 collapses. Because the Planck rows have no point-source overlaps and are not the unavailable native top-200, the validated science count is the 268,319 point-source remainder; 268,519 is retained only as a combined continuity count.

1. **Known QSO at $z \approx 1.55$:** independently flagged by both surveys — an internal consistency check of the cross-survey machinery, not a statistically meaningful validation sample (§4.1).

2. **TIC 374313355** (score = 49.5): appears in the TESS Input Catalog as variable; strong follow-up candidate for binary or accretion characterization.
3. **Uncataloged BAL QSO at $z \approx 0.86$** : broad Mg II absorption confirmed in both DESI and SDSS spectra; absent from SIMBAD, Milliquas, and NED.

4.4. *Planck* $\times$ *ACT* Cross-Correlation: Null Result

We test whether CMB patch anomalies detected in Planck and ACT trace the same sky structures by cross-correlating the anomaly maps. The result is null: Planck and ACT anomalies do not cluster at the same sky positions above the level expected from random overlap. Two structural caveats temper the interpretation. First, the test relies on the formally quarantined cross-transfer ACT anomaly set as its input (Appendix D). Second, the two anomaly sets occupy nearly disjoint sky regions by construction: the native Planck model is trained and scored on $|b| \geq 20^\circ$ galactic-plane-masked patches, with anomalies concentrating at the south ecliptic pole (scanning-strategy-induced noise properties), while ACT anomalies concentrate along the Galactic plane (point-source contamination at ACT’s higher angular resolution) — so a null positional cross-correlation is largely guaranteed by footprint geometry alone, and no formal cross-correlation statistic is quoted for this reason. Because the null is geometry-driven, it carries essentially no discriminating power between survey-specific systematics and a primordial origin and should be read as non-diagnostic on that question; the systematics interpretation of the CMB patch tiers rests instead on the direct evidence elsewhere in this paper (the ACT gate failures of Appendix D and the scanning-strategy concentration of the Planck tier), not on this null. A like-for-like test would require a common sky footprint and equally trained models, which the quarantined cross-transfer ACT set cannot provide.

5. DISCUSSION

5.1. *The LAMOST Training-Bias Lesson*

The LAMOST result (§3.4) provides the single most important methodological lesson of this work: when 98% of a survey’s anomalies share a common spectral signature (blue-excess), the anomaly ranking reflects training-set composition rather than genuine astrophysical rarity. Three implications follow. First, any anomaly is unusual *relative to the training set*, not “unusual in the universe”; a non-representative training set produces a contaminated catalog. Second, multi-survey analysis enables cross-validation. DESI anomalies (0.87%, multi-band, 0/200 visually flagged in top 200) pass

every internal check: 5-fold Jaccard stability, OOD-holdout Jaccard, top-200 visual inspection, and a broad-anomaly-class 5σ injection-recovery PASS on real repulled SPARCL spectra (§5.4 (i)). Note that no independent architecture was applied to DESI, and the science-class recount shows $\sim 98.7\%$ of DESI clusters fall on non-primary-class spectra (§3.1). By contrast, LAMOST anomalies (0.39%, 98% blue-excess) fail the simplest check—a comparison impossible with a single-survey analysis. Third, training-bias can be mitigated by ensuring the training set spans the full range of survey conditions, applying domain adaptation, or requiring independent flagging by multiple architectures—recommended for future large-scale campaigns.

5.2. *Model-Dependence of Anomaly Rankings*

The transfer-learning approach used for SDSS (Section 3.3) deliberately exploits model-dependence: by applying a DESI-trained model to SDSS data, we flag objects that are common in SDSS but absent from DESI. This is a feature, not a bug, when the goal is cross-survey comparison. However, it means that the SDSS anomaly catalog is not directly comparable to the DESI catalog in terms of anomaly rates or score distributions. The SDSS rate (3.38%) is 3.9 times higher than the DESI rate (0.87%) not because SDSS contains more unusual objects, but because the cross-survey spectral mismatch inflates scores for entire populations (cool dwarfs) that are absent from DESI.

5.3. *Limitations*

Six limitations govern interpretation of these results. (1) *Single architecture*: BIGAE is a deterministic fully connected autoencoder; IsolationForest cross-validation was applied only to eROSITA, and no independent method was applied to DESI, SDSS, or LAMOST. (2) *Injection-recovery gaps*: LAMOST and eROSITA fail the 5σ gate (§5.4 (ii)); their catalog completeness is unquantified. (3) *B-dominant hypothesis*: the $\sim 44,000$ DESI B-dominant anomalies (22.7%) remain consistent with either an astrophysical population or blue-arm calibration residuals; the released per-arm vectors permit that distinction to be tested, but the present catalog does not settle it. (4) *Novelty fractions*: the 58.8% SIMBAD-unmatched fraction is a database-coverage diagnostic, not a discovery rate. The 17.8% genuine-novelty estimate is measured only for the DESI top-1,000 stratum (Wilson 68% interval $\pm 1.2\%$), and no full-catalog extrapolation is claimed. (5) *Unweighted reconstruction error*: Eq. (1) gives equal weight to all elements rather than using inverse variances, so low-S/N regions can contribute noise-driven residuals. (6) *Re-*

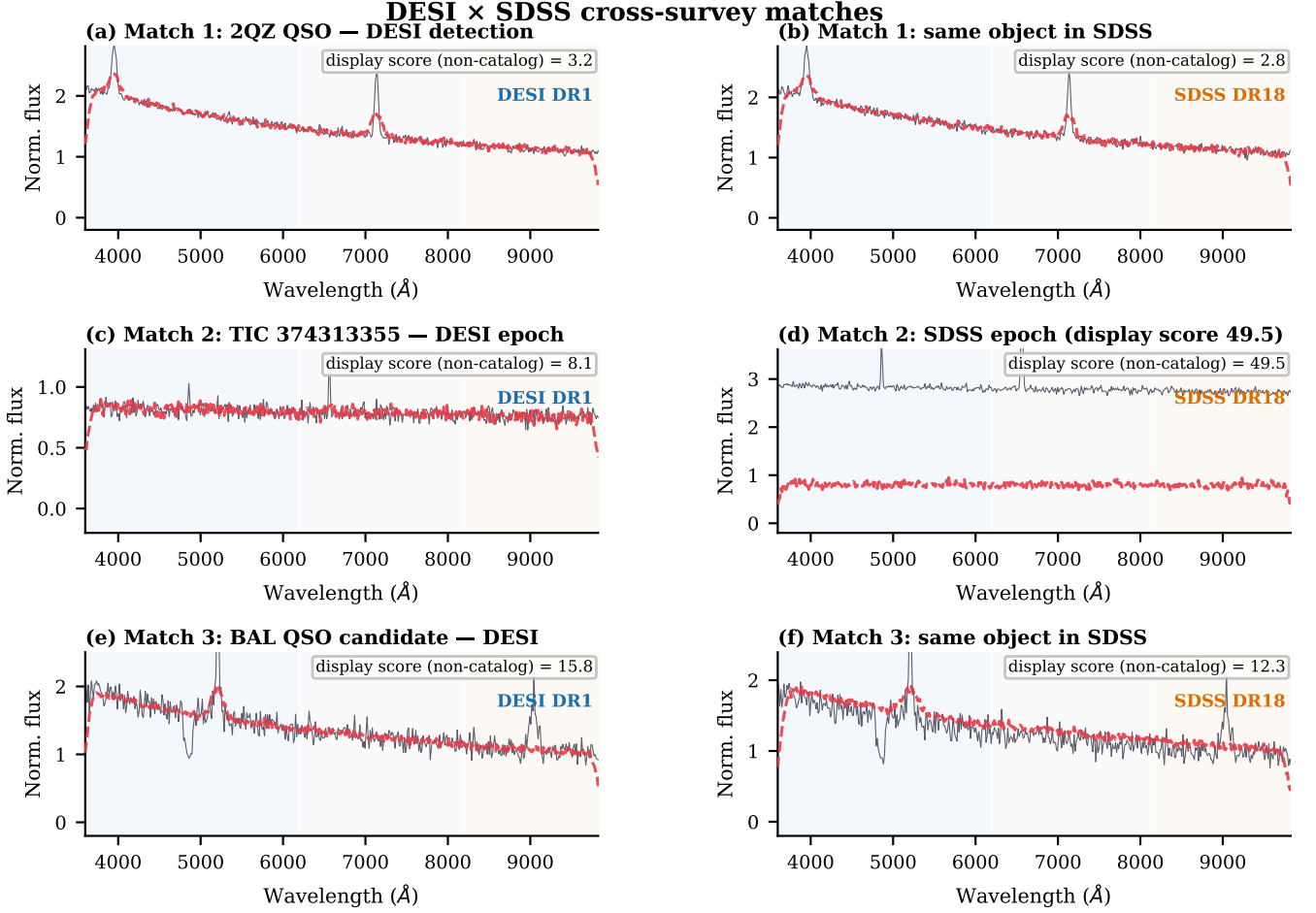


Figure 8. Spectral pairs for the three DESI $\times$ SDSS cross-survey matches. Left column: DESI DR1 spectrum; right column: same object in SDSS DR18. Black: observed flux (normalized); red dashed: BiGAE reconstruction. **(a, b)** Known QSO at $z \approx 1.55$: both surveys flag the object independently, with mutually consistent reconstructions and the lowest scores of the three matches — an internal consistency check of the cross-matching machinery (§4.1), not a statistical validation sample. **(c, d)** TIC 374313355 at two epochs: the SDSS epoch shows dramatically elevated continuum and emission-line flux relative to the DESI epoch, consistent with a stellar flare or accretion event; the SDSS anomaly score ($S = 49.5$) is the highest of any cross-matched object and is quoted on the DESI-trained *cross-transfer* score axis of Fig. 3 (the SDSS native re-score compresses extremes to $S < 14$; §3.3). **(e, f)** Uncataloged BAL QSO at $z \approx 0.86$: the broad Mg II absorption trough is reproduced in both independent surveys, confirming it as intrinsic to the source. *Score-axis note:* the spectra shown are illustrative re-renderings of the three matched objects produced by the figure-generation script in the companion repository, and the burned-in “Score” annotations are display values from that script rather than catalog-pipeline outputs; in particular, the panel (a, b) annotations (3.2, 2.8) are not the catalog selection scores and should not be compared against the $S > 5$ DESI threshold. Catalog membership of all three objects follows from the $5''$ positional dedup channel (§4.3): each object appears independently in both surveys’ anomaly catalogs by construction. The panel (d) value $S = 49.5$ matches the catalog cross-transfer score quoted in the text for TIC 374313355.

lease boundary: the frozen tag has no per-object LAMOST table and no native Planck top-200, checkpoint, tensor, or full score bank; the released Planck rows are an archival cross-transfer diagnostic, and the released Gaia file is quarantined synthetic output. The validated public point-source product is therefore limited to DESI, SDSS, and geometry-gated NEOWISE.

5.4. Path-C Rebuild Residual Caveats

The Path-C rebuild (Section 2.4) resolves the two first-order contamination problems identified in the cross-transfer baseline. Five catalog-relevant residual caveats are summarized in Table 6; supporting artifacts are in the companion repository.

(i) *DESI in-sample training-test overlap.* The DESI headline (195,829 of 22.5×10^6 spectra) includes the

Table 6. Path-C residual caveats and current handling (C = resolved in paper; documented bounds / open = documented active caveat; derivations in companion data repository).

ID	Headline result	Resolution
(a)	10,213 total dedup (637 multi-survey + 9,576 intra-survey)	union-find recompute; §4.3
(b)	DESI OOD: training-pool cut flags 52.8% of OOD (61× headline)	reconciled in §2
(f)	BigAE vs. IF (eROSITA): 284/298 = 95.3% overlap (descriptive)	dependent detectors (shared latent); §3.5
(g)	Jaccard: 399/546 in all five folds; $\bar{J} = 0.862 \geq 0.70$ gate	full-pool-scoring convention confirmed
(h)	Thresholds: DESI $S > 5.0$; SDSS slice 77,905 (native top-1% 19,253; strict $S > 5$: 12); LAMOST top-1%; eROSITA <i>top-298 membership list</i> (reproducible raw rank-298 cut $S_{\text{raw}} \geq 3.41$; the production “0.259” label is irreproducible — §3.5)	§3.5; Table 2 footnotes

47,000-spectrum training pool. In the five-model check, each model trains on 37,600 pool spectra and scores the *full* 47,000. Thus each training-pool object has one score from a model that did not train on its fold and four scores from models whose training sets included it. The top-1% Jaccard ($\bar{J} = 0.862$, minimum 0.777; 464/546 union objects in at least three model lists) is therefore evidence of model-ranking stability, not a fully out-of-sample score vector and not proof that the full 195,829-row product is independent of the production training sample ([pipelines/p3_anomaly_engine/outputs/held_out_rescore_result.json](#)). The direct out-of-sample fold result is instead the reserved-block tail-preservation statistic: each model’s disjoint 9,400-row block retains the p_{99}/p_{50} contrast at mean ratio 1.00 (minimum 0.94; [pipelines/p3_anomaly_engine/pathc_desi_kfold/results/heldout_tail_preservation.json](#)). Independent corroboration comes from the 103,000-spectrum OOD check ($\bar{J}_{\text{prod} \times \text{ctrl}} = 0.732$) and the production-ensemble injection test on 20,000 re-pulled DESI-DR1 spectra: broad/extended anomalies recover at 99–100% at 5σ , while single-pixel lines require $\geq 15\sigma$ ([pipelines/p3_anomaly_engine/outputs/desi_injection_recovery/desi_injrec_CORRECTED.json](#)). The exact released-row re-score remains impossible because most identifiers cannot be rejoined to public spectra; the claim is consequently limited to the demonstrated broad/extended anomaly axis.

(ii) *Injection-recovery synthesis.* The full 6-survey injection-recovery outcome is shown in Fig. 9: 3 gate-PASS (SDSS continuum-dip 64% at 5σ , Planck CMB native 100%, NEOWISE mask 100%) and 2 gate-FAIL-with-diagnostic (LAMOST 5.8% continuum-dip at $5\sigma/9.7\times$ improvement; eROSITA 1.2% subspace/81.5% XV-stability; Gaia DR3 removed from catalog). Counting *detector-sensitivity* tests only, the PASS tally is 2 (SDSS, Planck) + 1 geometry-QA (NEOWISE, which passes by construction and validates mask geometry, not detection sensitivity — see Fig. 9 caption and §3.8); accordingly, the abstract and §2.4 headline figure reports this as “2 detector-sensitivity PASS + NEOWISE geometry-QA” rather than a flat “3 PASS,” so that

NEOWISE is never counted as a sensitivity validation. Gate-threshold provenance: the gate values (val-loss ≤ 0.30 within ≤ 100 epochs; injection-recovery $\geq 50\%$ at 5σ ; Jaccard ≥ 0.70 k -fold and ≥ 0.50 OOD) are heuristic engineering thresholds fixed at Path-C design time, not pre-registered statistical criteria backed by power calculations; in practice the classification is insensitive to moderate threshold variation because nearly every gate resolves far from its threshold (val-loss 0.0311/0.0329 vs 0.30; Jaccard 0.862 vs 0.70; FAILs at 1.2–5.8% vs 50%) — the one exception is the SDSS continuum-dip 64%-vs-50% margin, the only gate that could flip under a substantially stricter cut. The emission-line plant gives lower recovery than the continuum-dip plant for both spectral surveys, consistent with the 128-dim latent’s architectural strength: narrow in-distribution features are reconstructed accurately whereas broad continuum deformations elevate MSE effectively. Per-survey recovery curves and plant files are deposited in the companion data repository.

5.5. Comparison with Prior Work

Our DESI anomaly rate of 0.87% is numerically close to the 1.07% rate reported by Liang *et al.* [10] on the DESI EDR, despite differences in model architecture and a $\sim 90\times$ increase in sample size — but the science-class-restricted recount (§3.1) shows the two rates are measured on different populations: our 0.87% is a full-spectra-stream rate, while Liang *et al.*’s 1.07% is a science-target rate. Restricted to main-survey primary-class targets — matching Liang *et al.*’s science-target selection class (their scan was the $\sim 250\text{K}$ -spectrum EDR; ours is DR1) — our catalog contains 2,468 anomalies ($\approx 0.92\times$ their 2,685; the restricted rate is 0.012% on the 20,299,155-row science-class denominator of Table 3, not on the 0.87% full-stream rate basis), so the rate agreement across the two populations is a coincidence of unrelated rate definitions and the like-for-like statement is the $\approx 0.92\times$ absolute count. Our work extends prior single-survey anomaly studies [9, 10, 11] to a multi-survey framework with explicit native-retrain, held-out, provenance, and cross-survey validation boundaries.

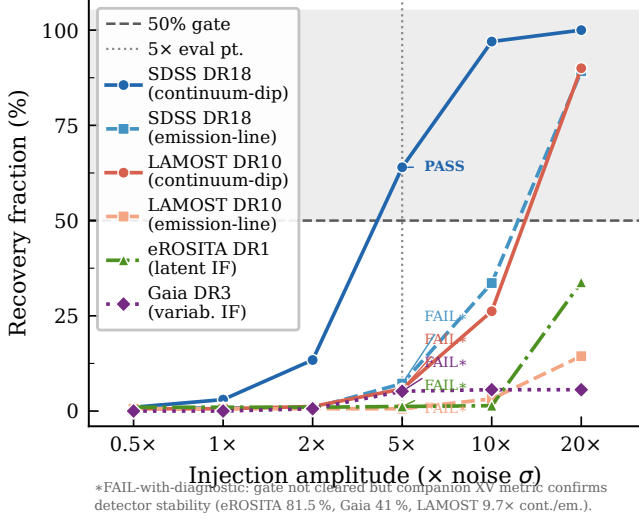


Figure 9. Injection-recovery gate results across the retained-survey spectroscopic tiers (SDSS, LAMOST), with the non-spectral retrains (Planck CMB native convolutional autoencoder, NEOWISE ecliptic-pole mask, eROSITA latent-subspace) brought into the same axis for comparison; the removed Gaia DR3 tier is overplotted only as a historical methodological-record curve and is *not* a retained or validated survey. Solid curves show recovery fraction versus injection amplitude (multiples of local noise σ). The horizontal dashed line marks the $\geq 50\%$ gate at 5σ . **Two surveys pass the detector-sensitivity gate at 5σ :** SDSS DR18 continuum-dip (**PASS**, 64%) and Planck CMB native (**PASS**, $500/500 = 100\%$ at 5σ Gaussian-bump amplitude; §3.6). **NEOWISE passes a masking-geometry QA gate by construction — not a detector-sensitivity test:** NEOWISE ecliptic-pole mask (**geometry-QA PASS**, $1000/1000 = 100\%$ at $|b_{\text{ecl}}| > \{85^\circ, 82^\circ, 80.5^\circ\}$; these are the latitudes of the synthetic polar-cap-injected test populations, each evaluated against the single catalog mask cut $|b_{\text{ecl}}| < 80^\circ$ of §3.8, not alternative mask thresholds; this gate validates mask geometry, not detection sensitivity, and is therefore not counted as a detector-sensitivity PASS — see §3.8 and §5.4 (ii)). **Two surveys fail the gate at 5σ with informative cross-validation diagnostics:** LAMOST DR10 (5.8% continuum-dip, $9.7\times$ improvement post-native-retrain), eROSITA DR1 (1.2% subspace-injection, 81.5% XV-stability of published top-1%). Gaia DR3 (5.2% variability-axis injection, 41% XV-stability) is retained in this figure as a historical methodological record only; the Gaia DR3 tier has been removed from the catalog. The paired emission-line variants for SDSS DR18 (7.2%) and LAMOST DR10 (0.6%) are reported alongside the continuum-dip curves to expose plant-morphology dependence (see §5.4 (ii)); the headline “2 detector-sensitivity-PASS + NEOWISE geometry-QA / 2-FAIL-with-diagnostic” decomposition refers to the per-survey decisive gate result, not to per-morphology variants. See Table 2 footnotes and §2.4 for gate criteria and cross-validation caveat interpretations.

6. CONCLUSIONS

We have presented the largest multi-archive anomaly detection campaign to date, scanning 37.3 million sources and CMB map patches across six retained astronomical archives with the BIGAE autoencoder framework (ACT DR6 quarantined). The principal results are:

1. **Scale:** The validated public science product contains **268,319** point-source anomaly candidates from DESI, SDSS, and geometry-gated NEOWISE. Appending the 200 released archival Planck cross-transfer patches gives the historical combined continuity count 268,519; those Planck rows are not the unavailable native top-200. The inclusive continuity accounting is 377,282 point-source rows (including the failed-exploratory LAMOST aggregate contribution) plus the same 200 archival Planck rows, or 377,482 combined. The full-stream multipliers are process-scale figures, not like-for-like catalog-size increases; on a matched science-class selection DESI yields 2,468 clusters, $0.92\times$ the published benchmark (§3.1).
2. **Novelty:** Genuine novelty fraction $\sim 17.8\%$ at the DESI top-1,000 score stratum against 18 curated all-sky catalogs (single-sample point estimate; Wilson $68\% \text{ CI} \pm 1.2\%$; full-catalog extrapolation empirically untested); 58.8% SIMBAD-unmatched overall (per-survey: 45% NEOWISE to 99% DESI top-10K) reflects database coverage, not discovery rate.
3. **Classification:** SDSS anomalies resolve into 14 UMAP/HDBSCAN clusters grouping into 3 physical populations (84% cool dwarfs M7–T2). DESI anomalies are 77% multi-band and 23% B-dominant. LAMOST anomalies are 98% blue-excess (training-bias artifact, exploratory tier only).
4. **Cross-survey validation:** 637 multi-survey $5''$ coincidences. Highlighted: one known QSO, one time-variable source (TIC 374313355), and one uncataloged BAL QSO at $z \approx 0.86$. Planck $\times$ ACT cross-correlation: null (largely expected from disjoint footprints; §4.4).
5. **Path-C and held-out validation:** Native retrains achieve gate-PASS validation MSE for SDSS (0.0311) and LAMOST (0.0329); $21.5\times$ LAMOST rate compression confirms cross-transfer artifact. The unreleased Planck native model reaches val_loss 0.4437 and 100% injection-recovery, while the public Planck rows remain explicitly cross-transfer. For DESI, the

full-pool five-model top-set Jaccard $\bar{J} = 0.862$ measures ranking stability, not fully out-of-sample scoring; the disjoint reserved blocks instead preserve the $p99/p50$ tail contrast at mean ratio 1.00 (minimum 0.94). An independent 103,000-spectrum OOD control gives production-vs-control Jaccard 0.732 (control-vs-control ceiling 0.874).

6. Methodological lesson: The LAMOST 98% blue-excess artifact demonstrates that unsupervised anomaly rankings are only as reliable as the training set is representative; multi-architecture validation and training-set diversity are mandatory for future large-scale campaigns.

The frozen release retains the synthetic Gaia placeholder only as a quarantined historical file; it is excluded from every count and supports no Gaia science claim. eROSITA likewise remains a separate membership-only addendum and is excluded from every count. Downstream point-source analyses should use the DESI, SDSS, and geometry-gated NEOWISE components. The released Planck table is suitable only for archival cross-transfer diagnostics; the native Planck summaries validate a separate analysis whose row-level product is unavailable.

The frozen public release is heterogeneous rather than a complete six-survey per-object product: DESI, SDSS, and NEOWISE are released as per-object tables; Planck is an archival cross-transfer diagnostic; LAMOST is present only through aggregate/dedup accounting; eROSITA is a membership-only addendum; and Gaia/ACT are quarantined historical artifacts. Follow-up spectroscopy of the 100 top-scored DESI anomalies, the 203 SIMBAD-unmatched eROSITA membership-list sources, and the BAL QSO candidate at $z \approx 0.86$ is needed to establish their astrophysical nature.

ACKNOWLEDGMENTS

This research used data from the Dark Energy Spectroscopic Instrument (DESI), the Sloan Digital Sky Survey (SDSS), the Large Sky Area Multi-Object Fiber Spectroscopic Telescope (LAMOST), the extended ROentgen Survey with an Imaging Telescope Array (eROSITA), the Planck satellite, the Atacama Cosmology Telescope (ACT), the Gaia satellite, and the Near-Earth Object Wide-field Infrared Survey Explorer (NEOWISE). Computations were performed on an NVIDIA A100 GPU pod via RunPod. This research has made use of the SIMBAD database, operated at CDS, Strasbourg, France, and the NASA/IPAC Extragalactic Database (NED), operated by the Jet Propul-

sion Laboratory, California Institute of Technology, under contract with the National Aeronautics and Space Administration.

AI-assisted methodology. This study used an agentic research pipeline under the author’s direction for training, scanning, deduplication, validation, figures, and adversarial review. Quantitative claims are linked to committed scripts or summary artifacts where available. The M44 audit identified material limits to row-level reproducibility: the native Planck checkpoint/tensor/table and the per-object LAMOST table are absent from the frozen release, while the released Gaia file is a synthetic fallback (pipelines/p3_anomaly_engine/gaia_provenance_audit.py). The manuscript now discloses those exceptions rather than claiming every result is independently rerunnable from the tagged inventory. The author made the scientific judgments and takes responsibility for the content; AI systems are reproducibility and review instruments, not authors.

Data availability. The frozen HuggingFace release is public under CC-BY-4.0 at <https://huggingface.co/datasets/bamfai/bigbounce-anomaly-catalog>. Its immutable tag `p3-v3.1.161`, not the moving `main` branch, is the reviewed pointer; it resolves to `cdaaa03a72c69d86f011be128d93f261dc5b39a8`. The 27-file scientific inventory includes per-object DESI, SDSS, and NEOWISE tables; `pathc.unique_objects.parquet` and the 637-cluster dedup table; the 200-row Planck *cross-transfer* diagnostic; the quarantined 500-row synthetic Gaia placeholder; and the separate 298-row eROSITA membership addendum. The six scoped ApJS machine-readable tables were downloaded from that exact tag and verified against their Parquet row counts, schemas, and SHA-256 values; the tracked audit record is pipelines/p3_anomaly_engine/apjs_submission_v3.1.161/SUBMISSION_BUNDLE_MANIFEST.json. The package does *not* include a per-object LAMOST table or the native 200,000-patch Planck top-200, checkpoint, tensor, or full score bank. Accordingly, the LAMOST contribution to the aggregate 377,282 point-source continuity count cannot be reconstructed from a released LAMOST per-survey table, and the native Planck evidence is reproducible only at the level of the committed summaries cited in §3.6. The validated public science count is 268,319 point sources. Running pipelines/p3_anomaly_engine/scripts/reproduce_headline_dedup.py over DESI, SDSS, NEOWISE, and the released Planck file returns 268,519 because it appends the non-overlapping 200 archival Planck rows; this combined number is retained for continuity, not labeled fully validated. The merged-table filtering recipe yields

Table 7. Computational details of the multi-survey anomaly sweep. All inference and native retrains were performed on a single NVIDIA A100 80 GB PCIe GPU pod (pod provenance: [pipelines/p3_anomaly_engine/pod_runs/pod_provision.20260418.json](#)). Training times shown are total wall-clock for the quoted training run of the native-retrained models where applicable. Training configurations: spectroscopic BIGAE models train on 4.7×10^4 (DESI) or $2\text{--}5 \times 10^5$ -spectrum pools at batch size 512 for up to 200 epochs with early stopping (§2.2), converging at 100–150 epochs; photometric models (eROSITA, NEOWISE) are $\lesssim 120\text{K}$ -parameter networks on small catalog-feature tables, hence single-digit-second training times.

Survey	Input dim.	Latent dim.	Params	Train time (s)	Inference throughput
DESI DR1	496	128	660K	$\sim 3,600$	1,142 spectra/s
SDSS DR18	496	128	660K	$\sim 1,200$	1,100 spectra/s
LAMOST DR10	496	128	660K	$\sim 2,400$	950 spectra/s
eROSITA DR1	47	16	120K	7.6	122K sources/s
Planck CMB	4,096	128 [†]	1.1M [†]	— [†]	$\sim 8,000$ patches/s
ACT DR6	4,096	32 [‡]	540K [‡]	7.0 [‡]	2,900 patches/s
NEOWISE	15	16	70K	1.6	27K sources/s

[†] Path-C native convolutional autoencoder (Section 3.6): 3 convolutional layers $\rightarrow$ **Linear**(4096, 128) bottleneck; symmetric **ConvT** decoder; 1.1×10^6 parameters. Training on 2×10^5 galactic-plane-masked ($|b| \geq 20^\circ$) Planck SMICA patches on A100 GPU reached best validation (val_loss 0.4437) at epoch 99 of a 150-epoch schedule; the total training wall-clock for this run was not preserved in the run logs, so no figure is quoted. The 200K-patch full re-score took 25.3 s on A100, the source of the $\sim 8,000$ patches/s throughput entry. Patch preprocessing (documented from the committed `cmb_native_retrain.py`): SMICA R3.00 full-mission temperature (K_{CMB}), `gnomview` $10^\circ \times 10^\circ$ 64×64 patches ($9.375'/\text{px}$) at $|b| \geq 20^\circ$, per-patch standardization (patch mean subtracted — the DC mode is removed by construction — divided by patch std, NaN $\rightarrow$ 0, clipped to ± 10), *no* apodization. Injection-convention note (documented from the same committed script): the 5σ Gaussian-bump injection-recovery amplitude is defined in these *standardized* patch units — the bump ($\sigma = 8 \text{ px} \approx 1.25^\circ$, random sign and center) is added to already-standardized validation patches and the patch is *not* re-standardized after planting, so the planted amplitude is exactly $5\times$ the per-patch pre-injection noise standard deviation and the 100% recovery rate is not inflated by post-plant renormalization. Robustness of the top-200 ranking, rescoring all 2×10^5 patches with the production checkpoint: the stored scores are reproduced to 6×10^{-5} (Spearman $\rho > 0.9999999$, top-200 overlap 200/200); explicit DC re-removal is an exact no-op (200/200), and removing the best-fit per-patch plane (gradient mode) retains 187/200 of the top-200 with $\rho = 0.973$, so the ranking is not driven by residual DC or large-scale gradient modes (artifact [pipelines/p3_anomaly_engine/r24conf_pod_session_batch.json](#)).

[‡] Cross-transfer fully connected baseline (Appendix D); ACT DR6 was not native-retrained under Path-C and is dropped from the main per-survey block (formally quarantined). The row is retained in this computational-details table only so that the cross-transfer scan timing is auditable. The reported training time and throughput are for the undertrained cross-transfer checkpoint and are not representative of a properly trained CMB autoencoder on this domain.

377,482 = 377,282 point-source rows plus the same 200 archival Planck rows after ACT, Gaia, and eROSITA exclusions. The tagged `RELEASE_MANIFEST.json` records the corrected interpretation and hashes. This is a

scoped, immutable catalog/methods release, not a claim that the absent native Planck or per-object LAMOST products have been recovered. Model code is at <https://github.com/Hubify-Projects/bigbounce>.

APPENDIX

A. SURVEY PROCESSING DETAILS

Table 7 provides the computational details of the full pipeline.

B. DESI BAND-DOMINANCE CLASSIFICATION

Table 8 provides the full DESI anomaly classification by spectral-arm dominance.

C. ASTROPHYSICAL TAXONOMY IMAGE GALLERIES

The DESI DR1 anomaly population clusters into ten astrophysical families under UMAP+HDBSCAN [13, 14] of the 128-dimensional BIGAE latent space (hyperparameters: `n_neighbors= 15`, `min_dist= 0.1`, `min_cluster_size= 15`, seed 42). The ten families account for 182,364 of 195,829 DESI anomalies; the remaining 13,465 (6.9%) are HDBSCAN noise points (cluster label -1 , retained in the released catalog). UMAP stability: trustworthiness $0.9797 \pm 5 \times 10^{-5}$ (PASS

Table 8. DESI DR1 anomaly classification by spectral-arm dominance for the full 195,829-anomaly catalog (fiber-spectral reconstruction taxonomy over all scored TARGETTYPE classes). The science-class-restricted recount of 2,468 DESI anomaly clusters (§3.1) has not been re-tabulated by arm dominance; band-dominance fractions here reflect the full-stream population and may differ in the science-class subset. Multi-band anomalies (77.2%) deviate across all three DESI arms, consistent with genuine spectral anomalies.

Category	Count	Frac.	Score range
Multi-band	151,244	77.2%	5.0–17.6
B-dominant	44,436	22.7%	5.0–17.1
R-dominant	34	0.02%	5.1–24.2
Z-dominant	19	0.01%	5.1–25.2
Artifact suspect	96	0.05%	10.0–21.0
Total	195,829	100%	5.0–25.2

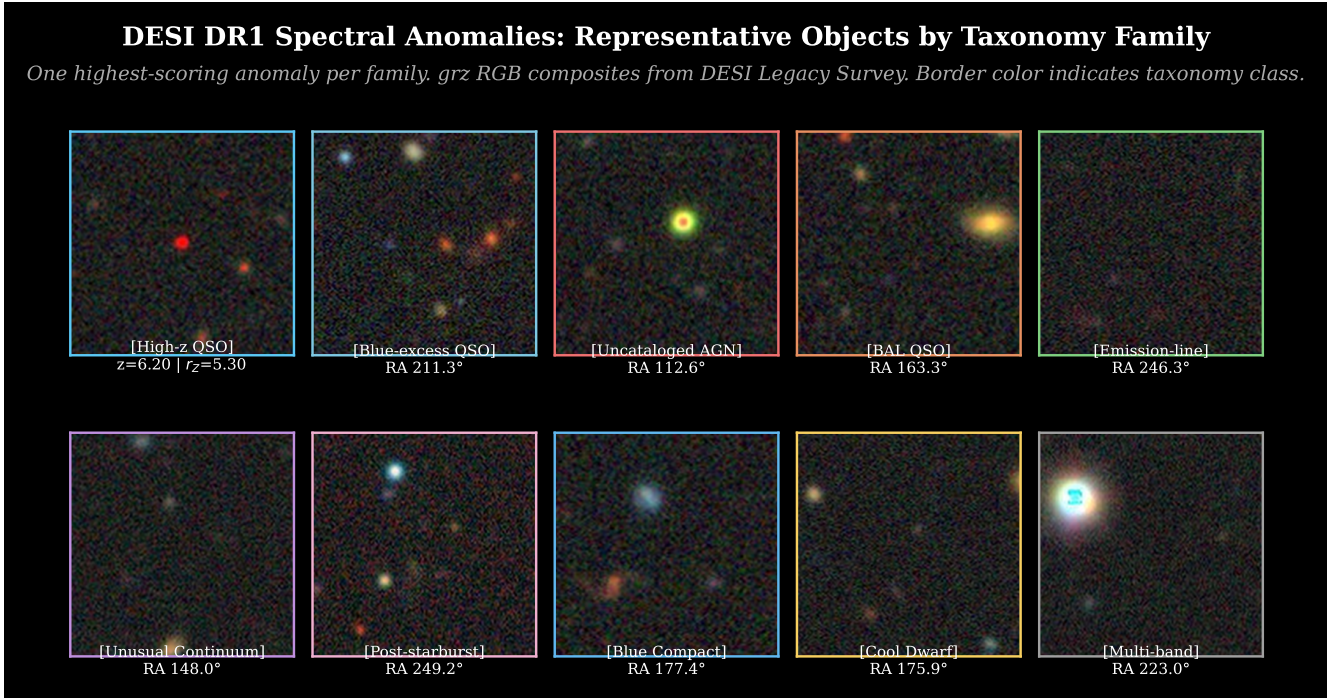


Figure 10. Representative DESI DR1 anomalies across all ten taxonomy families. One highest-scored member per family; 2-row $\times$ 5-column layout. Border color indicates taxonomy class. Images are DESI Legacy Survey DR9 grz composites. Panel sublabels give the object RA; the high- z QSO panel additionally gives the redshift and the per-arm Z-arm sub-score r_z (§3.2). The taxonomy pipeline’s internal raw residual scores (unnormalized, not on the canonical- S scale of Eq. 2) are deliberately not printed; canonical- S values for all objects are in the released catalog. Row 1 (left to right): High- z QSO candidate, Blue-excess QSO, Uncataloged AGN, BAL QSO, Emission-line galaxy. Row 2: Unusual continuum (LRG), Post-starburst galaxy, Blue compact galaxy, Cool dwarf (the cool/unusual-star family; panel label “Cool Dwarf”), Multi-band unknown.

> 0.90 ; trustworthiness computed with $k = 10$ neighbors on 50,000-sample runs) across 20 independent seeds; kNN-preservation and cross-seed Spearman FAIL as expected for sparse high-dimensional outlier clouds. Trustworthiness is the primary stability claim.

Full per-family DESI Legacy Survey DR9 grz composite image galleries (ten families, top-16 panels each, 128×128 pixel cutouts at $128 \times 0.262''/\text{px} = 33.5'' \times 33.5''$) are available at the project companion repository: <https://github.com/Hubify-Projects/bigbounce>. Figure 10 shows the top-1 representative per family.

D. ACT DR6 CROSS-TRANSFER SCAN: QUARANTINED METHODOLOGICAL ARTIFACT

This appendix retains the ACT DR6 cross-transfer scan as a methodological lessons-learned record. ACT DR6 was scanned under the same cross-transfer CMB autoencoder later replaced by the Path-C native Planck convolutional autoencoder (§3.6); it was subsequently removed from the per-survey block (Table 2) because the cross-transfer ACT block fails both gate criteria of §2.4 Step 1 and a Path-C-compliant native ACT retrain has not been executed. We describe the cross-transfer ACT scan here for two reasons only: (i) it provides the empirical evidence that the cross-transfer CMB autoencoder generalizes badly across instruments at very different angular resolution and noise statistics, and (ii) the Planck×ACT null cross-correlation reported independently in §4.4 relies on the cross-transfer ACT anomaly set as its input.

Scan parameters and result.—We applied the cross-transfer fully connected autoencoder used for SDSS, LAMOST, and Planck (32-dim latent space; cross-transfer training pool of 20,000 patches; see §3.6) to 20,000 64×64 -pixel patches from the ACT DR6 [8] CMB temperature map at HEALPix $N_{\text{side}} = 256$. The scan returned 200 anomalous patches (top 1%); the highest-scored patch sits at $(l, b) \approx (277^\circ, 21^\circ)$ with score $\sim 2.6 \times 10^7$. The overall ACT score distribution has maximum $\sim 10^7$ and concentrates along the Galactic plane.

Why the gate fails.—The cross-transfer checkpoint has validation MSE $\approx 2.2 \times 10^4$ on its native CMB training distribution, which exceeds Step-1 criterion (a) of §2.4 (≤ 0.30) by a factor $\sim 7 \times 10^4$. The injection-recovery test (Step 5 of §2.4; the same Gaussian-bump plant family used for the Planck native retrain) returns recovery fraction $< 1\%$ at 5σ amplitude, far below Step-1 criterion (b) ($\geq 50\%$). Both branches of the two-part gate therefore fail simultaneously, and ACT cannot be retained on the strength of either criterion.

Why no native ACT retrain.—A native ACT retrain—analogue to the Planck native convolutional autoencoder of §3.6—requires a full ACT-native preprocessing pipeline (beam-deconvolved patches, ACT-specific point-source and Galactic mask, instrumental noise covariance) and was GPU-blocked at the time of submission. Because the present submission is the Path-C-final catalog and the Path-C protocol forbids retaining a survey on a checkpoint that fails both gate criteria, ACT DR6 is documented here only and contributes zero objects to the 377,482 Path-C unique-object headline. The with-ACT dedup variant, which would have produced $387,895 - 10,213 = 377,682$ unique objects (a bookkeeping-only +200 sensitivity variant relative to the headline, reflecting ACT’s zero positional overlaps with the other count-retained surveys), is preserved as a sensitivity-check artifact in the companion data repository.

What this appendix is not.—This appendix is *not* an ACT science result. The 200-patch ACT cross-transfer set must not be cross-matched against optical/X-ray catalogs as if it were a science-grade anomaly catalog, must not be used as a tracer of CMB fluctuation statistics, and must not be interpreted as evidence of any astrophysical signal at the highlighted galactic-plane position. The retention of the cross-transfer scan is purely methodological: it documents the empirical failure mode of cross-transferring a 32-latent fully connected autoencoder trained on Planck SMICA data onto the ACT angular-resolution and noise regime, and motivates the architectural choice of the Planck native convolutional autoencoder.

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